

Vayu VY-6

Flight Deck and Human Factors Design Description

VAY-HF-0001 · Issue 1 · 2 September 2026

Companion to VAY-SDD-0001, Architecture and System Design Description, Issue 1

DEMONSTRATION ARTICLE. The Vayu VY-6 is a representative six-passenger electric vertical take-off and landing air taxi constructed by Safety Lab Aero. Aircraft-level characteristics follow published information for aircraft in this class and are listed against their sources in Appendix B of VAY-SDD-0001. The system architecture below aircraft level, the flight deck, and every crew task in this document are representative, and are not the design data of any manufacturer.

This document describes the design and stops where the assessment begins. Section 11 states what it therefore does not contain, and why each omission is deliberate.

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1 Purpose and scope

VAY-SDD-0001 describes what each of the aircraft's sixteen systems is, what it draws on, and where it sits in the airframe. It says almost nothing about the one interface every one of those systems ultimately terminates at: the single person on the flight deck.

This document supplies that missing description. It is the design input to the human-factors assessment — function allocation, crew task analysis, human error analysis, crew alerting, ergonomics, situation awareness, the controls-and-displays evaluation, and the minimum-flight-crew determination for the type.

It follows the same rule as the SDD: **it describes the design, and it stops where the assessment begins**. §11 states precisely what it therefore does not contain, and why that omission is deliberate.

A word on why this document exists at all for a single-pilot aircraft. It would be easy to assume that a one-person flight deck needs less human-factors design description than a two-person one, on the theory that there is no crew coordination to describe. The opposite is closer to true. Every item that a two-crew flight deck can hand to the other seat — a cross-check, a second set of eyes on an alert, a division of the radio and the checklist — has to be absorbed by the flight deck itself on this aircraft, or it does not happen. That makes the geometry, the display content, the control layout and the alerting philosophy carry more of the burden here, not less, and it is why this document is written with the same care as its two-crew sibling rather than a lighter one.

The aircraft this document describes is divided in VAY-SDD-0001 into sixteen systems across ten zones, and every one of those systems eventually reaches the flight deck in some form — a display page, a lever, a switch, an annunciator, or simply a design choice about what the pilot is never shown at all. This document does not repeat the system descriptions; VAY-SDD-0001 does that. It extracts, from each of the sixteen, the fact that lands on the one person who has to fly, navigate, communicate, and manage the aircraft's power and thermal state without another qualified person to share any part of it with. Where the SDD is organised by system, this document is organised by what the pilot sees, touches, and is told, and it draws its material from the SDD's own words wherever a crew-relevant fact is stated there.

This document also does not repeat the aircraft-level configuration data in VAY-SDD-0001 §1.2 — span, mass, propulsor count, cruise speed and the rest — except where a figure has a direct bearing on the crew interface, such as a sector length that sets the phase durations in §2.2, or a passenger count that sets the cabin accommodation this document has to account for. Readers wanting the aircraft's full configuration should go to the SDD directly.

The certification basis is likewise a matter for VAY-SDD-0001 §1.3 rather than for this document: the intended basis is a special condition for small-category vertical take-off and landing aircraft, with the applicable category left open pending the intended operation and the maximum seating configuration. This document does not name a paragraph of that basis, does not assume which category applies, and does not anticipate what either choice would require of the flight deck. Where the analyses this document feeds need a specific evaluation method or a specific numbered requirement, that method and that requirement are theirs to bring, not this document's to guess at.

1.1 Provenance convention

Tag	Meaning
[GIVEN]	Fixed by the aircraft specification or by VAY-SDD-0001. Do not expect it to move.
[PRELIM]	Preliminary design data. Real, but not frozen. Cite as preliminary.
[ASSUMED]	Engineering assumption, recorded so it can be challenged. Not given to us.
[TBD]	A known gap, named rather than filled.

Where a value the analyses need does not yet exist it is marked [TBD] and left empty. **An empty field is the correct answer.** A plausible number substituted for a missing one is worse than the gap, because the gap is visible and the substitution is not.

2 Operating concept, as the crew experiences it

The mission characteristics in VAY-SDD-0001 §1.1 have consequences for the crew that the SDD does not draw out. They are stated here as facts, not as findings.

Characteristic	Value	Provenance
Sector length	20 – 30 km	[GIVEN]
Sector rate	8 – 14 per day per airframe	[GIVEN]
Flight rules	Day and night operation stated; the applicable flight rules are not	[TBD]
Cruise altitude	300 – 600 m above the city	[GIVEN]
Crew complement carried	1 pilot. No cabin crew.	[GIVEN]
Passengers carried	6, in the cabin (zone 200)	[GIVEN]
Boarding / deboarding	A ground operation at the pad, aircraft powered, pilot remaining in the seat throughout	[GIVEN]
Turnaround	Approximately 10 minutes	[GIVEN]
Line maintenance at the pad	None	[GIVEN]
Energy replenishment	Available only at defined network points, not at every pad	[GIVEN]
Pad environment	Elevated structures surrounded by buildings; shear and turbulence off structures a normal condition	[GIVEN]

These are stated in VAY-SDD-0001 §1.1 as mission characteristics, not as flight-deck design; this document’s job is to say what each one means once it reaches the pilot. A sector rate of eight to fourteen a day, for instance, is an airframe utilisation figure in the SDD. For the pilot it is eight to fourteen repetitions, in a single duty period, of every task this document describes: a boarding check with no second person to share it, a vertical departure, a transition, a cruise, a second transition, an approach into a building-shear environment, and a landing, each with its own display and alerting behaviour. Day and night operation means that repetition happens under two different ambient-lighting regimes, sometimes within the same duty period, against pads whose own lighting and surrounding-structure visibility is not characterised in either document (§10.2).

2.1 Three consequences worth stating plainly

There is no second person. Whatever a two-crew flight deck hands across the centre pedestal to the other seat — a cross-check of a setting, a second reading of an alert, a shared radio and checklist workload — this flight deck has to absorb by itself, because there is nobody else on it. [GIVEN] This is not a statement about whether one pilot is enough; §3 says plainly that the determination of minimum flight crew is not made here. It is a statement about what the flight-deck design has to be capable of regardless of that determination’s outcome: every display page, every alert, and every control in the sections that follow has exactly one possible reader, and the design either accounts for that or it does not.

The turnaround is a powered, occupied event, not a shutdown. The aircraft does not go quiet between legs. It stays powered, the pilot stays in the seat, and passengers board and leave around a live aircraft close to a building edge, roughly ten minutes at a time, eight to fourteen times a day. [GIVEN] Ten minutes is not idle time from the flight deck’s point of view. In that window the pilot is, at minimum, the person confirming doors and restraints before the next departure (§7.3, §8.4), and possibly also monitoring energy replenishment where a pad has that capability, all without having left the seat or handed any part of the task to someone else on the flight deck.

Every sector carries two transitions, not the occasional one. The aircraft has no tilting mechanism; the change between hover-borne and wing-borne flight is a matter of control law and airspeed, executed once outbound and once inbound on every single leg. Across a 14-sector day that is up to 28 transitions for one aircraft, far more frequent than the occasional regime change a conventional or tilt-rotor type would present. [GIVEN] Section 9 records that the transition sequence itself is commanded automatically by the flight control system once conditions permit; what stays with the pilot, every time, is flying the aircraft through it and reading whatever the displays and alerting system show while it happens.

2.2 Flight phases

The programme phase vocabulary, built from the representative sector in VAY-SDD-0001 §1.1. Durations are the nominal split used for design purposes; they are not a schedule. [PRELIM]

Phase	Nominal duration (min)
Standing / pre-flight	[TBD]
Boarding	about 10 (turnaround figure)
Vertical departure (hover, climb)	2
Transition to wing-borne flight	Included in departure/climb above
Wing-borne cruise	9
Transition to hover-borne flight	Included in approach below
Approach and landing (hover, descent)	3
Deboarding	Included in turnaround above

The 2-minute departure and 3-minute approach figures each contain a transition; VAY-SDD-0001 does not break the transition itself out as a separately timed sub-phase, and no attempt is made here to invent one. [GIVEN] for the named minutes; [TBD] for a standalone transition duration and for standing time.

The resulting sector shape is short and front- and back-loaded: nine minutes of steady wing-borne cruise sits between two shorter, denser phases — the vertical departure and climb, and the approach and landing — each of which folds a hover regime, a transition, and (on the approach side) a building-shaded pad environment into two or three minutes. A phase table built this way is a design input for the task and workload analyses to divide further; it is not itself a claim about how much happens inside either short phase, which is exactly the kind of question this document leaves to those analyses.

3 Crew station and qualification

Item	Value	Provenance
Pilot stations	1, on the aircraft centreline	[GIVEN]
Additional flight-deck seating	None described	[GIVEN]
Passenger seating	6, cabin (zone 200), two-two-two, aft of the flight deck	[GIVEN]
Cabin crew	None	[GIVEN]
Oxygen provision	None described in VAY-SDD-0001	[GIVEN]
Operating convention	Single continuous pilot duty; no Pilot Flying / Pilot Monitoring exchange is possible	[GIVEN]
Kind of operation sought	Day and night	[GIVEN]

The absence of an oxygen system is stated here as a design fact rather than an oversight: cruise altitude is 300–600 m above the city, well below where crew oxygen is normally a design feature, and VAY-SDD-0001's sixteen-system inventory names no such system. If that changes, this table changes with it.

The single-station arrangement also removes a category of design question that a two-seat flight deck has to answer and this one does not: there is no question of which seat is primary, no handover of controls between seats, and no seat-to-seat reach or sightline problem to solve, because there is only one seat. What replaces those questions is a different one — whether everything the pilot needs across a full sector, from boarding check to shutdown, is reachable, visible and legible from that one position without having to rely on a second viewpoint that does not exist. Sections 4 through 8 are the design description that question is asked of; §11 restates why answering it is not this document's job.

The one-pilot figure is an input, not an answer. VAY-SDD-0001 gives a crew complement of one pilot and no cabin crew. That is the specification's sizing assumption. The minimum-flight-crew determination for the type is made by the analysis against the workload methodology applicable to the

certification basis, and may confirm the number, or challenge it. Nothing in this document performs that determination.

The “no additional flight-deck seating” row carries a training consequence worth stating alongside it: VAY-SDD-0001 describes no jumpseat or observer position on the flight deck, which means a check pilot, an instructor, or an observer cannot occupy the flight deck during a revenue sector to watch a line pilot fly the type as flown. Whatever line-observation the training and checking programme relies on has to happen some other way — from the cabin, from the simulator named in §3.1, or not as direct observation at all. Which of those applies is not stated and is [TBD].

3.1 Qualification and training [ASSUMED]

- Type-rated pilots, current on type, drawn from a pool flying repetitive short-sector urban operations at high daily tempo.
- Recurrent training is assumed to include the elevated-pad approach in a building-shear environment, the hover/wing-borne transition in each direction, and the no-cabin-crew boarding and passenger-readiness check as named items.
- A flight-deck mock-up and a full-flight simulator are planned programme assets. No simulator campaign has been run. [TBD] — evidence dates.
- Because there is no second pilot to develop line experience alongside, the qualification route by which a newly type-rated pilot is judged ready for unsupervised single-pilot line flying — line-check content, sign-off authority, minimum experience before solo operation — is [TBD].
- A daily tempo of eight to fourteen sectors, flown alone, by day and by night, is itself a training consideration this document flags without resolving: whatever recency and currency requirements the operator sets, the pilot accrues them at a much higher repetition rate per duty day than a lower-tempo type would produce, and does so without a second pilot present to observe or moderate the day. What recency and duty-time provisions apply is an operating-rule question outside this document’s scope. [TBD]
- Recurrent evaluation of the pilot’s handling of the flight deck’s automatic behaviours — the automatic transition sequencing, the automatic passenger-address announcements, and the automatic detect-and-avoid track build described in §9 — as distinct, named training items is [TBD].

4 Flight deck architecture and geometry

The flight deck occupies zone 100, on the aircraft centreline, forward of the cabin. Per VAY-SDD-0001’s zone table, the boundary at the front of the cabin is a pressure-free bulkhead, and the flight deck is **open to the cabin above seat height** rather than separated from it by a door. [GIVEN]

4.1 Consequences of the single-pilot centreline arrangement [PRELIM]

- Seating is symmetric on the centreline rather than offset to a side, so the eye-reference-point question is one of longitudinal and vertical position only, not of a left/right pairing as on a side-by-side flight deck.
- Every flight-deck resource that a two-person aircraft can split across two seats — one display each, one radio each, one set of eyes on the alert list each — exists here in a single instance for the pilot to use alone, whatever redundancy the underlying system carries. Section 6 records where the display and control architecture is itself duplicated; that duplication serves system availability, and its relationship to a single user’s workload is for the analyses.
- The flight deck being open to the cabin above seat height means the pilot has a direct, unobstructed sightline over the passenger seats. VAY-SDD-0001 does not describe a door, a partition, or a means of closing that opening. What that means for cabin noise reaching the flight deck, for passenger access to the flight deck in flight, and for any security or isolation function, is not addressed by either document and is [TBD].
- That same open boundary is also the only means by which the pilot can directly observe the cabin without a camera or other sensor — none is described in VAY-SDD-0001. Section 7.3 records that passenger-readiness confirmation before departure falls to the pilot or ground staff with no cabin crew present; whether that confirmation in practice relies on this sightline, on a display indication, or on a ground signal is [TBD] there.
- Boarding takes place with the flight deck occupied and the aircraft powered, which is the normal condition for every one of the eight to fourteen daily turnarounds rather than an exceptional one. The pilot conducts whatever pre-departure checks the type requires from the seat, without a second flight-deck occupant to perform or verify any part of them.

- The centreline position also fixes, by construction, where the geometry problems in §4.3 have to be solved: there is one Design Eye Position to define, not two, and every panel zone in §4.2 is described relative to that single position rather than to a left- or right-hand offset. That simplifies the geometry problem in one sense and removes any possibility of compensating for a poor solution from a second vantage point in another.
- The urban pad environment described in §2 — elevated structures, buildings close in, shear and turbulence off nearby surfaces as a normal condition — is flown into and out of on every sector by the same one pilot occupying the same fixed centreline position, using whatever external vision envelope item 2 of §4.3 eventually defines. A wider or narrower cut-off angle than a conventional transport's has direct consequence for how much of the pad edge, the surrounding structures, and any obstacle close to the aircraft is visible during the two- to three-minute departure and approach phases, independent of anything the displays in §6 show.

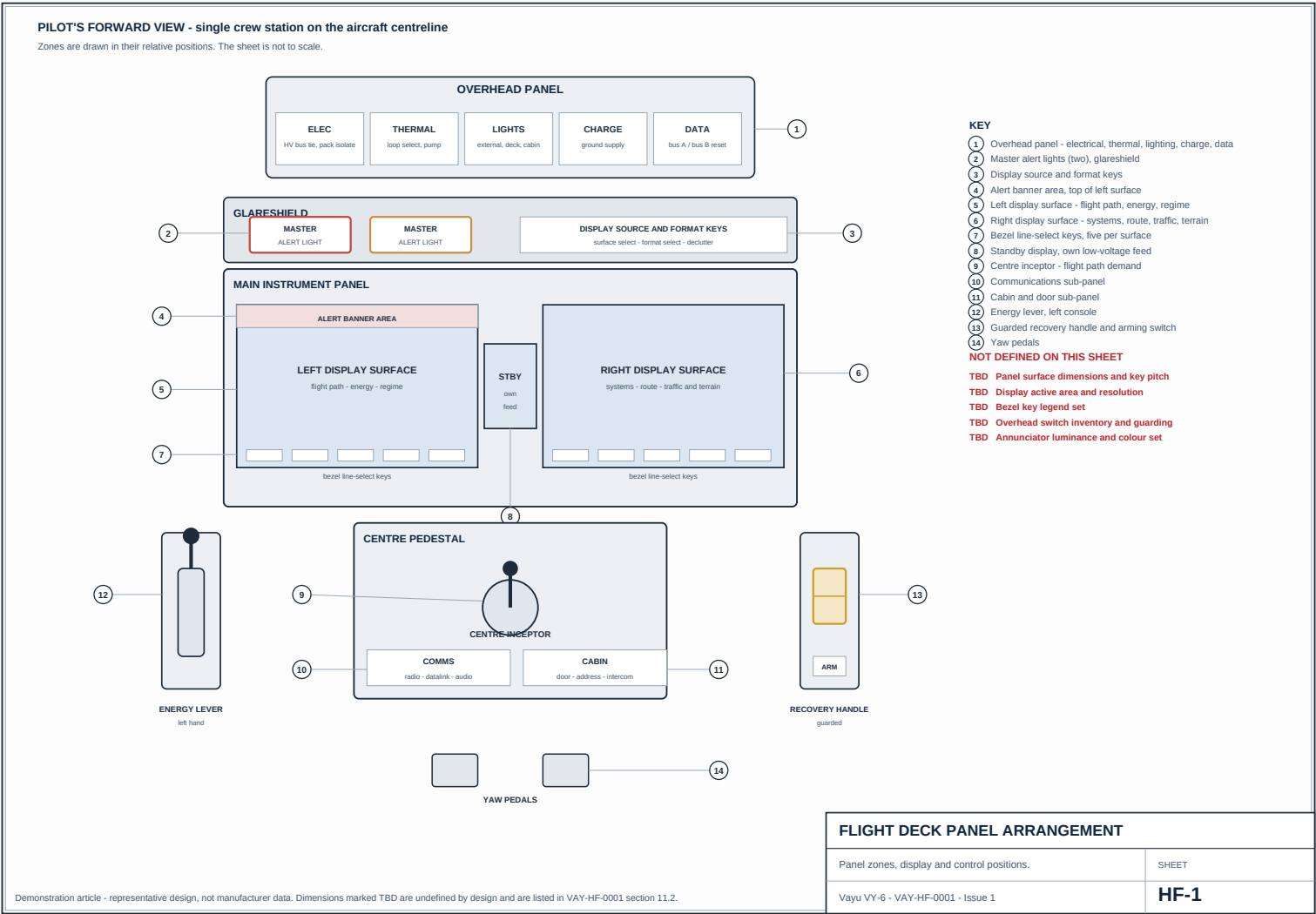


Figure HF-1 — Flight deck panel arrangement. Panel zones, display and control positions [PRELIM]. Also issued as a standalone drawing sheet.

4.2 Panel zones [PRELIM]

VAY-SDD-0001 contains no dedicated flight-deck layout drawing; the panel-zone table below is assembled from the equipment placements each of the sixteen system descriptions states in passing — DIS locating its own displays and switches, FCS and NAV locating their computers, ERS locating its handle and switch, and so on. That method of assembly is stated plainly because it means this table is only as complete as the placements the SDD happens to mention, and a zone with no equipment listed against it is not evidence that the zone is empty, only that nothing in the SDD’s text places anything there yet.

Zone	Contents
Alerting panel (above the displays)	Master caution and warning lights; annunciator legends
Main instrument panel	Two display surfaces, each on its own graphics processor and configurable to flight, navigation or synoptic pages; a standby display; an audio panel
Lower instrument panel	Hard switches for gear, lighting, and system resets
Centre console	Centre inceptor (flight path); left-hand energy lever; guarded emergency-recovery handle and its separate safe-and-arm switch
Dedicated annunciator	Emergency recovery system control-unit status (armed state, fault) — see §8.3
Overhead panel	[TBD] — not described in VAY-SDD-0001
Side consoles	[TBD] — not described in VAY-SDD-0001

Two things are worth stating plainly about this table. First, VAY-SDD-0001 describes a lower-instrument-panel location for the hard switches that a conventional two-crew transport commonly places overhead; whether that is because this aircraft genuinely has no overhead panel, or because the overhead panel simply has not been described yet, is itself [TBD]. Second, the emergency-recovery annunciator is called out separately from the alerting panel because VAY-SDD-0001 describes it as “a dedicated flight deck annunciator near the activation handle” rather than as part of the general alert list — see §8.3 for what that distinction means for the crew interface.

A third point follows from the first two together. Where a conventional two-crew flight deck spreads switches, annunciators and displays across an overhead panel, a main panel and two side consoles reachable from two different seats, this aircraft’s described equipment concentrates into three areas immediately in front of a single seat: the alerting panel above, the displays and standby display in the centre, and the lower panel and console below and beside. Whether that concentration is complete — whether an overhead panel and side consoles genuinely do not exist, or simply have not yet been placed in the SDD’s text — is exactly the kind of question §4.3’s geometry items and this table’s [TBD] rows exist to carry forward rather than answer by assumption.

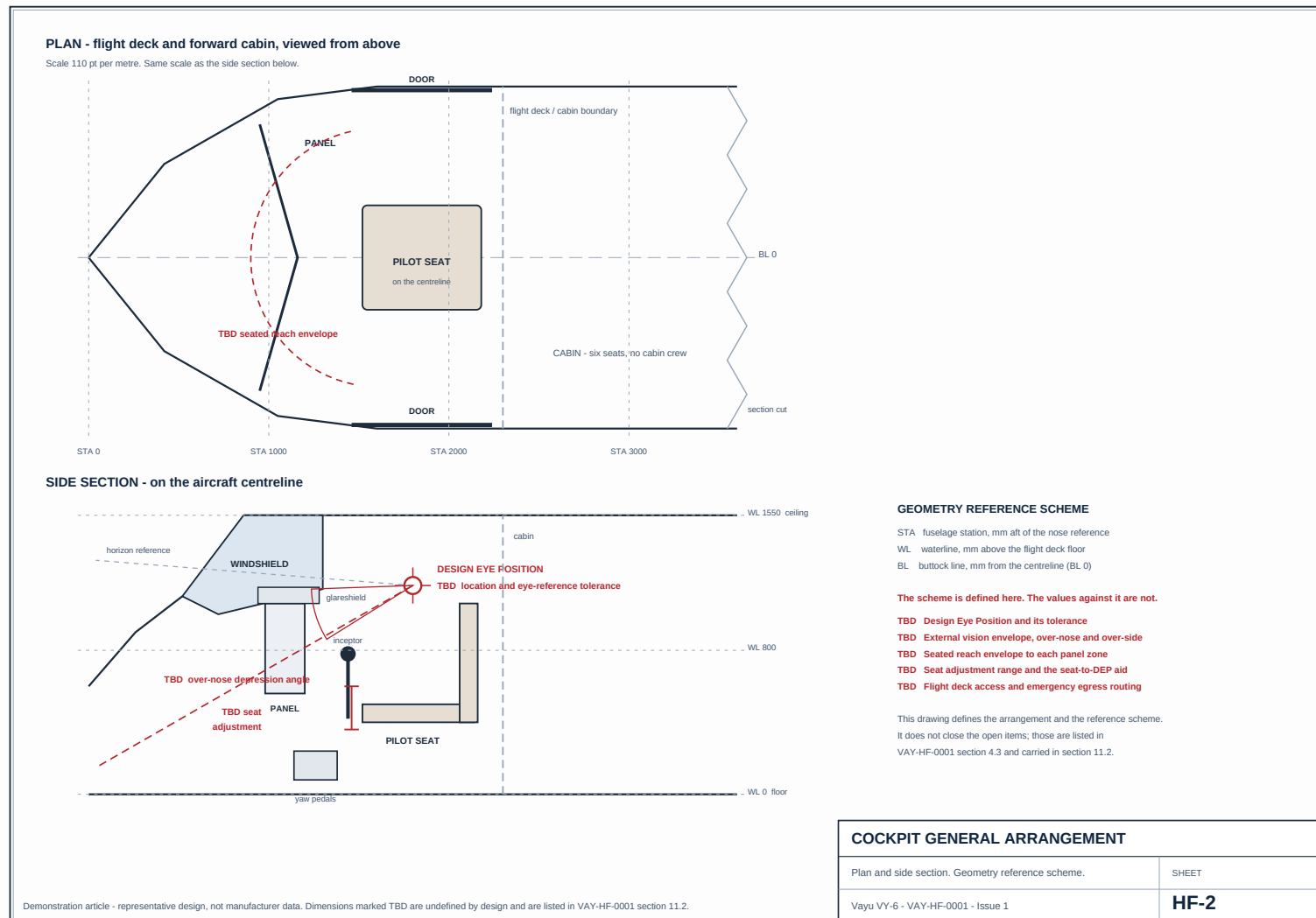


Figure HF-2 — Cockpit general arrangement, plan and side section. The drawing defines the geometry reference scheme; the values against it are the open items of §4.3. Also issued as a standalone drawing sheet.

4.3 Geometry data [TBD]

None of the six items below exist yet in any form this document can cite. They are listed because the ergonomics and controls-and-displays analyses cannot proceed without them, and because a single-seat flight deck has no second position from which a poor answer to any one of them could be partially compensated for. Required, and **not yet defined**:

1. Design Eye Position and eye-reference-point tolerance
2. External vision envelope — over-nose and over-side cut-off angles, with particular attention to close-quarters visibility of the pad, its edge, and the surrounding building structures during an elevated-pad approach
3. Sightline and viewing angle from the design eye position to each display surface and to the standby display
4. Reach envelope to each panel zone from the seat design position
5. Seat adjustment range and the seat-to-eye-reference alignment aid
6. Flight-deck access and emergency egress routing, including how the open boundary to the cabin figures in an egress path

5 Anthropometry and accommodation

Item	Value	Provenance
Accommodation range	5th percentile female to 95th percentile male	[ASSUMED]
Anthropometric database	[TBD] — not selected	
Clothing / equipment state	Normal flight clothing; no cold-weather or high-visibility bulk case defined	[TBD]
Demonstration method	Mock-up, digital human model, or both	[TBD]

With one pilot seat and no second position to compensate for a poor fit at the first, the accommodation range has to be validated across its full span rather than checked at a nominal midpoint; a geometry that is merely adequate for a central percentile is not enough when there is no second pilot able to reach what the first cannot. That is a design intent stated for the accommodation work to carry forward, not a finding that any current geometry meets it.

The accommodation range interacts directly with the geometry items §4.3 lists as [TBD]: a Design Eye Position, a seat adjustment range, and a reach envelope are each only meaningful once the accommodation range they are meant to cover is fixed. Selecting the anthropometric database and the demonstration method is accordingly a precondition for closing those geometry items, not a parallel and independent task. Passenger anthropometry — six variably-sized occupants per leg, seated aft of the flight deck with restraints described in §7.3 and §8.4 — is a separate question from crew accommodation and is not addressed by this section; nothing in VAY-SDD-0001 or this document's source material sets a passenger anthropometric basis either.

6 Display suite

Per VAY-SDD-0001 §5.11 (DIS). [GIVEN] unless noted.

Element	Provision
Display surfaces	2, side by side, each on its own graphics processor, configurable to flight, navigation or synoptic pages through bezel keys
Standby display	1, per the aircraft configuration. Its architecture is not detailed in the DIS system description. [TBD]
Format flexibility	Either display can show any available page, including the other surface's page.

Element	Provision
Flight-path input	Centre inceptor, with two independent signal paths to the two flight control computers
Energy input	Left-hand energy lever, sending independent position signals to each flight control computer
Reference data	1 inertial measurement unit and 1 satellite positioning receiver (zone 100); forward air data probes on the nose; low-speed sensors near each boom root, reporting independently per boom
Alerting hardware	Master caution and warning lights and annunciator legends on a panel above the displays; an aural tone generator sharing the COM audio-panel path

6.1 Reconfiguration behaviour [GIVEN] where stated, [TBD] where not

- **Page selection between the two display surfaces is manual**, made by the pilot at the bezel keys; either surface can be set to show any available page.
- **What happens automatically, if anything, on the loss of one display's graphics processor is not stated.** Whether the surviving surface takes up the lost content on its own, whether the pilot must select it, or whether some other reversion path exists, is [TBD].
- The navigation and air-data sensor sets are described in the singular for inertial and satellite positioning — one unit each — with only the low-speed sensors described as present at both booms. Whether the primary flight displays have any means of reversion if the single inertial or satellite positioning source is affected is [TBD].

That combination — manual page selection as the normal way of working, paired with an unstated reversion behaviour on processor loss, and a navigation sensor set that is single rather than duplicated where the displays themselves are — is recorded here as design fact. What it means for the crew is for the analyses.

One further mechanical point is worth recording plainly, because it follows directly from there being two independently selectable surfaces and one reader. At any moment, the pilot's two displays can show a primary flight page and a second page together, two non-flight pages together, or the same page twice; VAY-SDD-0001 states that either surface can carry either format, but does not state that a primary flight format is protected from being displaced off both surfaces at once. Whether such protection exists, in software or in procedure, is [TBD] and is listed again at item 8 below.

6.2 Display content [PRELIM]

Content named in VAY-SDD-0001 as reaching the displays, drawn from the DIS system description and from the interfaces other systems declare to DIS:

- Flight, navigation and system synoptic pages, selectable to either surface
- Traffic and terrain tracks from DAA, on a dedicated page monitored during pad approach and departure
- A power/energy page: bus voltage, contactor state and isolation resistance (from EPD); battery charge and health (from HVB); coolant temperature per loop, and battery and inverter temperatures (from TMS)
- COM channel and datalink status, on a system synoptic page
- Gear position, shown on the same page during extension and retraction (from LGS)
- A cabin synoptic page showing door and latch status (from CAB)
- Alert messages, appearing on whichever display shows a synoptic or status page, with an aural tone and one of the two glareshield alert lights. Which light accompanies which message is not defined [TBD]

Read across the sixteen system descriptions, the content list above is not a fixed set of pages so much as a statement that every system with something to report sends it to the same pair of surfaces: propulsion, energy, thermal, navigation, communications, traffic, gear, and cabin state all converge on the two displays and the standby display, selected and arranged by the one person reading them. VAY-SDD-0001 does not state a maximum number of pages, a paging hierarchy, or a rule for what happens when two systems need attention on two different pages at once; those are exactly the questions item 8 below leaves open.

6.3 Not defined [TBD]

The four items below are the ones the display-suite description above repeatedly points at without resolving. They matter more on this type than on a lower-tempo one because whatever the answer turns out to be, the pilot lives inside it eight to fourteen times a day, day and night, alone:

7. Measured luminance, contrast and sunlight-readability performance
 8. Which format each display shows by default, per phase
 9. Behaviour on loss of one, then both, display graphics processors
 10. The standby display's architecture — independence of its supply, sensor and processing path from the two primary displays
-

7 Controls

[PRELIM] unless noted.

7.1 Flight path and energy

- A single centre inceptor for flight path, with two independent signal paths carried to the two flight control computers [GIVEN]
- A single left-hand energy lever, likewise sending independent position signals to each flight control computer [GIVEN]
- A guarded emergency-recovery handle at the centre console, and a separate safe-and-arm switch: both are required before a firing signal can reach the recovery-system charge [GIVEN]
- Neither VAY-SDD-0001 nor this document's source material describes rudder pedals, a yaw pedal, or any other directional-control input distinct from the centre inceptor. Whether such an input exists, and if so what it drives, is [TBD].
- No autopilot engage/disengage control, autothrottle control, or flight-director selector is described — see §9.

There is no mechanical or hydraulic linkage anywhere between the flight deck and a propulsor or a control surface. Every command from the centre inceptor and the energy lever reaches its destination electrically, through the flight control computers, over the aircraft's two data buses. This document records that architecture; it does not analyse the single-point character of a one-lever, one-inceptor input path.

The flight-path and energy inputs are worth setting against the corresponding arrangement on a larger, multi-propulsor transport. Where a four-engine aircraft gives the crew one thrust lever per engine, this aircraft gives the pilot one energy lever for all ten propulsors combined — eight lift rotors and two cruise propellers — with the flight control system, not the lever, deciding how commanded energy is distributed across whichever propulsors are active for the current flight regime. That is a direct consequence of the distributed-electric-propulsion architecture DEP and FCS describe in VAY-SDD-0001 §5: allocation across ten independent units is a control-law function, not something a pilot could do lever by lever even with more hands than one.

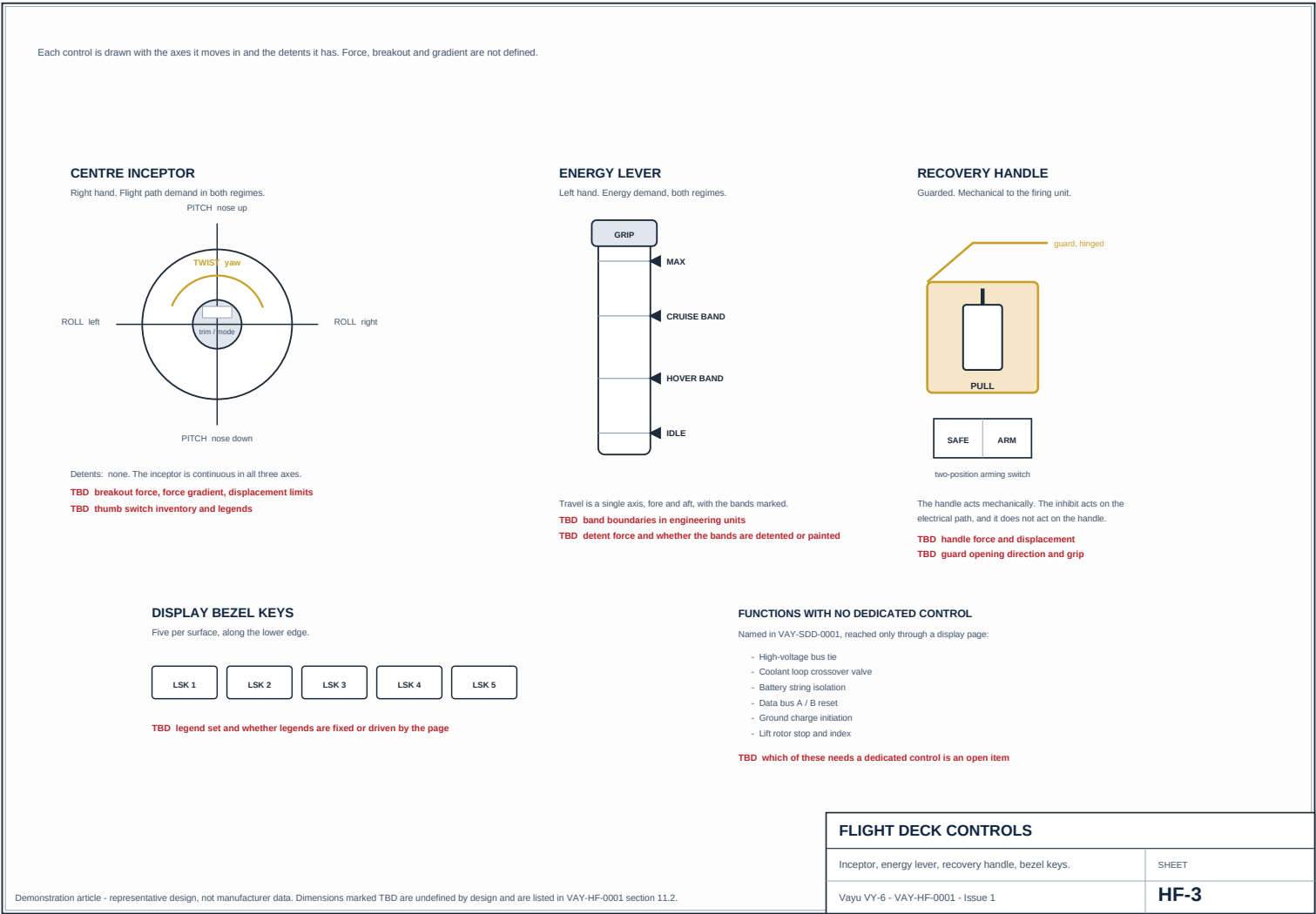


Figure HF-3 — Flight deck controls. Axes, detents and guarding [PRELIM]; forces and gradients are not defined. Also issued as a standalone drawing sheet.

7.2 Systems

Hard switches for landing gear, lighting and system resets are described at the lower instrument panel rather than overhead. [GIVEN] No overhead panel is described in VAY-SDD-0001; whether the type has one, and what philosophy governs it if so, is [TBD].

Beyond those named hard switches, VAY-SDD-0001 gives no account of how the pilot commands the remaining systems that plainly need some crew-accessible control at some point in a flight or a turnaround: closing or opening the electrical bus-tie contactor, selecting a coolant-loop configuration, initiating ground charging, or resetting a data-bus channel are each mentioned as things the systems themselves or the flight control system do, without a stated flight-deck control surface for a pilot override or manual selection where one might exist. Whether such controls exist and where they sit is [TBD], distinct from the hard-switch set named above.

7.3 Cabin and door controls

- Two cabin doors, one each side, each with an independent electric latch actuator and its own manual release lever [GIVEN]
- Door and latch status is presented to the pilot on a cabin synoptic page [GIVEN]
- Cabin lighting and the forward/aft environmental outlets are adjustable by the pilot from the flight deck [GIVEN]
- On arrival, the pilot confirms the aircraft is at the pad before releasing the door latches [GIVEN]. The flight-deck control by which that release is commanded — a dedicated door-control panel, a function on an existing panel, or something else — is not named in VAY-SDD-0001. [TBD]
- With no cabin crew carried, the pilot or ground staff confirm each passenger is seated and restrained before doors are requested closed [GIVEN]. How that confirmation is made — by the pilot’s own sightline over the open flight-deck/cabin boundary described in §4.1, by ground staff signal, by some display indication, or by more than one of these — is [TBD].
- Restraint buckle status is recorded by HMS “where sensed” [GIVEN], a qualifier VAY-SDD-0001 uses without stating which seat positions are instrumented and which are not. Whether every one of the six passenger restraints reports status, or only some, is [TBD].
- Cabin lighting is arranged as two strings, one per sidewall, on separate low-voltage feed segments, with ceiling trim lighting providing overlap [GIVEN]; the pilot’s control for selecting or overriding either string individually, as opposed to cabin lighting as a whole, is [TBD].

7.4 Control-actuation data [TBD]

As with the geometry items in §4.3, none of the following four items exist yet in a form this document can cite, and each is a precondition for the reach, force and error-tolerant-design analyses that the controls in §7.1 through §7.3 will eventually need to pass through:

11. Breakout forces, control travel, and detent forces on the centre inceptor, the energy lever, and the guarded recovery handle
12. Actuation direction conventions
13. Guarding and locking provisions for irreversible actions, beyond the recovery handle’s guard and its separate safe-and-arm switch
14. Presence and form of any yaw or directional control input separate from the centre inceptor

8 Crew alerting

This section describes the alerting system’s **capability**. It does **not** assign a priority to any condition — that population is an analysis output.

8.1 Capability [GIVEN] unless noted

Property	Provision
Hardware	An alerting panel above the two displays carries master caution and warning lights and annunciator legends

Property	Provision
Presentation	Alert messages appear on whichever display currently shows a synoptic or status page, with an aural tone and one of the two glareshield alert lights. Which light accompanies which message is [TBD]
Acknowledgement	A glareshield switch; the pilot can then call up the associated system page
Procedures	Whether the associated page called up on acknowledgement contains the alert's procedure, or only its system status, is [TBD]
Aural	A single tone generator, sharing the same audio path as the COM audio panel used for radio and intercom [GIVEN]. Precedence and arbitration between an alert tone and radio or intercom audio on that shared path is [TBD]
Tactile	[TBD] — no tactile alerting means is stated

The acknowledgement switch itself is placed at the glareshield [GIVEN], directly below the alerting panel it answers and within the same forward reach envelope as the displays it lets the pilot call a system page up from; whether that reach is confirmed for the accommodation range in §5, from the centreline seat, at the design eye position §4.3 has not yet fixed, is not established by either document.

The shared aural path is worth stating plainly: on this aircraft the same audio chain that carries an alert tone to the pilot's ear also carries every radio call and every cabin intercom exchange. Whether the two are arbitrated, and how, is not described.

That sharing is a direct consequence of the flight deck's single-occupant architecture rather than an oversight specific to alerting: the audio panel, the alert tone generator, the vertiport datalink, the two VHF radios and the cabin intercom all terminate at the one headset the pilot wears, because there is only one headset to terminate at. A two-crew flight deck can route different audio to each headset, or rely on one pilot to monitor the radio while the other attends to an alert; neither option exists here by construction, and this document records the single shared path as the design fact it is rather than assuming an arbitration scheme that is not stated.

8.2 Philosophy [ASSUMED]

- Alerts ordered by urgency of required crew response, not by system of origin
- One condition, one alert; the consolidation rule where several systems report related status — DAA's dual processing channels, for instance, or EPD's two bus sections — is [TBD]
- Phase-dependent inhibits during a vertical departure, a transition, or a pad approach: windows and inhibited set [TBD]
- Whether the alerting system distinguishes at all between a condition affecting the aircraft's ability to complete the current sector and a condition affecting only the next one — relevant on a type that returns to a chargeable node only at defined network points rather than at every pad — is [TBD]
- Whether any alert is suppressed, modified or re-timed specifically because the aircraft is on a pad, powered, with passengers boarding or leaving around it, as distinct from being suppressed for reasons tied to a conventional ground phase, is [TBD]

8.3 Indications that are not on a display [GIVEN]

- **Emergency-recovery system control-unit status** — armed state and fault — is presented on “a dedicated flight deck annunciator near the activation handle,” not on a display page. That distinguishes it from the general alert list in the same way a discrete hardware indication always does: it does not depend on which page either display currently shows. What VAY-SDD-0001 does **not** state is whether this annunciator's supply, sensing or processing path is independent of the two primary displays' own failure modes — unlike a purely mechanical indicator, it is described as presented through DIS, and its independence from a display or avionics failure is [TBD].
- **Master caution and warning lights** are named as hardware on the alerting panel, distinct from the display surfaces beneath them, in the same sense that any dedicated annunciator light is equipment rather than a reconfigurable page.

8.4 Two conditions the SDD singles out [GIVEN]

Stated as design facts. Their classification and their crew response are for the analyses:

- Cabin door and latch status is presented on a dedicated synoptic page, described specifically as being consulted “during boarding and before departure” — a condition the SDD calls out for crew attention at those two points in the turnaround.
 - Restraint buckle status is recorded “where sensed” rather than for every seat without qualification, which VAY-SDD-0001 states as a property of the recording rather than as an oversight to be corrected here.
-

9 Automation

[PRELIM] unless noted.

No autopilot, autothrottle, or flight-director function is described anywhere in VAY-SDD-0001. [GIVEN] The flight control system instead provides continuous fly-by-wire control laws — a hover law, a wing-borne law, and a transition law that blends the two — translating the pilot’s centre-inceptor and energy-lever inputs into propulsor and surface commands throughout the flight. [GIVEN]

- The transition itself is sequenced by the flight control system rather than by a discrete pilot selection: it commands the lift-rotor stop-and-align mechanisms “once conditions permit,” shifting authority from rotor thrust to aerodynamic surfaces as airspeed builds, and the reverse on the way back into hover. [GIVEN]
- Passenger-address announcements at boarding and on arrival are sequenced automatically from flight-phase information, without pilot input, with the pilot able to override from the flight-deck microphone. [GIVEN]
- Traffic and obstacle tracking runs continuously from engine start to shutdown; no pilot action is required to keep the track list current. [GIVEN]
- Thermal management pump speed is modulated automatically from reported heat load in each coolant loop; this is a system-level automatic function with no described crew-facing control. [GIVEN]
- Ground and air behaviour selection in the flight control system is driven automatically by weight-on-wheels state reported from the main landing-gear legs, with no described pilot selection between the two. [GIVEN]
- Detect-and-avoid’s two independent processing channels cross-check their track lists automatically before the result reaches the displays, and the processor falls back to the local sensor picture alone if the transponder path is affected, without a stated pilot action to trigger that fallback. [GIVEN]

Energy state feeds this picture continuously rather than at fixed checkpoints: HVB’s charge and health data “are displayed to the crew and used by FCS for energy planning,” and TMS’s per-loop coolant, battery and inverter temperatures are displayed alongside it. [GIVEN] Against an operating concept where a charge is available only at defined network points rather than at every pad (§2), that continuous display is the pilot’s stated window into whether the aircraft’s stored energy supports the sector ahead and the return to a chargeable point beyond it; no dispatch-level energy threshold, display cue, or decision point is named in either document, and its absence is tracked with the other procedural gaps at §10.1.

Set against those automatic functions, several closely related actions are described as remaining with the pilot rather than the system: gear retraction and extension are pilot-commanded, observed leg by leg on the display as each one cycles, with a manual extension handle available if the electric actuators do not respond; the safe-and-arm switch for the emergency recovery system is pilot-set according to phase of flight; and page selection between the two displays (§6.1) is manual throughout. The pattern that results — some functions automatic by default with no stated pilot involvement, others manual by default with no stated automatic assistance, on the same flight deck — is recorded here as a design fact in the same spirit as §6.1’s display finding. What it implies for the pilot’s model of what the aircraft is doing for them at a given moment, unassisted, is for the analyses.

Not defined [TBD], and each a precondition for understanding what the automatic functions above actually do at their edges:

15. Mode annunciation set, if any function beyond continuous fly-by-wire control exists; transition and reversion annunciation
16. Envelope protection — whether provided, and in which axes, in hover and in wing-borne flight
17. Any automatic reversion or disconnect condition for the fly-by-wire control laws, and its alerting
18. Control-law and transition-logic behaviour on loss of a display processor, one coolant loop, or one propulsor

19. Whether control gains or the transition threshold schedule on aircraft mass or loading — six passengers of varying weight is not a fixed item, and the aircraft's inertia distribution changes leg to leg with who is on board
-

10 Procedures, environment and crew-relevant interfaces

10.1 Procedures [TBD] throughout

VAY-SDD-0001 contains no procedures. A normal operating sequence, phase by phase, and a non-normal set organised by system, have been constructed for this document from the architecture, and appear as Appendix A and Appendix B. They are written as design input to the task, error and workload analyses rather than as an approved operating procedure, and they are marked throughout with the provenance of each step. Where a step would need a value nobody has defined, the step names the gap instead: recording an absence plainly is preferred to inventing a plausible-looking number that has not been designed.

Appendix C lists every control and indication on the flight deck against its panel zone and its provenance.

26. Backup medium for procedures — paper, electronic, or none, is not defined
27. Memory (recall) item set
28. Whether the system page called up on alert acknowledgement (§8.1) contains the associated procedure, or only system status
29. The boarding handover and cross-check between ground staff and the pilot, with no cabin crew present to share it — who confirms what, and in what order, before doors are requested closed

One procedural fact is given rather than open: the turnaround readiness report compiled by HMS at engine shutdown is reviewed by ground crew at a maintenance terminal before the aircraft is released for its next flight. [GIVEN] What the pilot's own part in that release decision is — beyond calling up recent recorded events on a maintenance page if needed — is not stated.

Ground charging is a further turnaround activity VAY-SDD-0001 places partly on the pilot's side of the interface: EPD reconfigures contactors to route charge current from ground equipment to each pack “under FCS supervision,” and each pack's controller accepts charge current independently. [GIVEN] Whether that supervision requires a pilot-initiated step, runs automatically once ground equipment connects, or something between the two, is [TBD]; it matters here because charging, where it happens, shares the same roughly ten-minute turnaround window as boarding, restraint confirmation and the readiness-report review, all without a second flight-deck occupant to divide the window between tasks.

The take-off, or vertical-departure, configuration question that a conventional transport answers with a dedicated configuration monitor is not addressed at all in VAY-SDD-0001: no such monitor, and no stated parameter set for one, appears anywhere in the sixteen system descriptions. Whether a pre-departure configuration check exists in any form, and if so what it covers, is [TBD] and is tracked with the other procedural gaps above.

10.2 Environment [TBD] throughout

None of the six items below has a measured value behind it anywhere in the source material this document draws on. They are grouped together because each describes the physical environment the pilot works inside rather than something the aircraft's own systems report, and because on this type — an open flight deck, an urban pad environment, and a duty day built from many short, low-altitude legs rather than few long ones — the environment is not a background condition so much as a recurring element of every single sector:

20. Ambient noise — in hover with all eight lift rotors active, in wing-borne cruise with the two propellers, and at the open flight-deck/cabin boundary described in §4.1
21. Vibration from the boom-mounted lift rotors and the wing-mounted cruise propellers, and its transmission to the flight deck
22. Lighting levels and independence of lighting supplies, across day and night operation
23. Wind, shear and turbulence loading at elevated pads among buildings — a stated operating condition (§2) with no measured data behind it
24. Visual environment on a night approach to an elevated, built-up pad
25. Electromagnetic and optical interference from adjacent structures, cranes and other rotorcraft in the urban corridor

Each of these is a design case named in general terms by VAY-SDD-0001 §1.1 — the pad environment “shaped by buildings rather than by the free-stream,” with “shear and turbulence off nearby structures” as “a normal operating condition rather than an exception” — without a measured value behind any of them. The gap matters more for this aircraft than it would for one flying a stable enroute environment for most of its sector, because on the VY-6 the building-shaped environment coincides with the phases where the pilot’s attention is already most divided: the vertical departure and the approach and landing, each carrying a transition, each only two to three minutes long, and each flown by one person with the full display and alerting picture from §6 and §8 to read at the same time.

10.3 Crew-relevant system characteristics

From VAY-SDD-0001. The characteristic listed is the one that reaches the crew. The table draws one row from each of the sixteen systems named in §1’s inventory, so that every system contributes something the pilot sees, uses, or is told, even where that contribution is narrower than the system’s full scope in the SDD; readers wanting the complete description of any one system should go to VAY-SDD-0001 §5 directly rather than treat this row as a summary of it.

System	Characteristic	Provenance
DEP	Ten independent inverter-motor units, one per propulsor, no gearbox; the flight control system redistributes torque across the survivors on the loss of any one	[GIVEN]
LFT	Eight lift rotors, each with its own stop-and-align mechanism; blades lock edge-on to the airflow in cruise	[GIVEN]
HVB	Two independent battery packs of four strings each; charge and health data displayed to the crew continuously	[GIVEN]
EPD	Two high-voltage bus sections joined by a tie contactor; bus voltage, contactor state and isolation resistance displayed to the crew	[GIVEN]
TMS	Two independent coolant loops, left and right; coolant temperature per loop, and battery and inverter temperatures, displayed to the crew	[GIVEN]
FCS	No autopilot, autothrottle or flight director described; continuous fly-by-wire control laws for hover, wing-borne flight and transition, commanded over two independent data buses	[GIVEN]
ACT	Electromechanical actuation of the ailerons and V-tail surfaces; no mechanical or hydraulic linkage to the flight deck; position and status reported back to the flight control system and the displays	[GIVEN]
NAV	A single inertial measurement unit and a single satellite positioning receiver; forward air data probes; low-speed sensors reporting independently at each boom	[GIVEN]
COM	Two VHF radios and one vertiport datalink transceiver; passenger-address announcements sequenced automatically with pilot override	[GIVEN]

System	Characteristic	Provenance
DAA	Nose, wingtip and tail sensors plus a transponder; two independent processing channels, cross-checked, building the traffic and terrain picture automatically	[GIVEN]
DIS	Two reconfigurable display surfaces plus a standby display; master caution and warning lights on a dedicated panel; a glareshield acknowledgement switch	[GIVEN]
LGS	A retractable, wheeled, four-point gear; weight-on-wheels sensed at each main leg; a mechanical downlock at each leg independent of continued actuator power; a manual extension handle in the flight deck	[GIVEN]
STR	Composite semi-monocoque structure; ground crew inspect gear-bay, battery-bay and boom access panels during turnaround — the pilot's part in that inspection, if any, is not stated	[GIVEN], [TBD] for pilot involvement
CAB	Six seats, two-two-two, each with a four-point restraint; two doors, each independently latched; no cabin crew; environmental and lighting control from the flight deck	[GIVEN]
ERS	A guarded activation handle and a separate safe-and-arm switch, both required to fire; a dedicated firing battery independent of the aircraft's main low-voltage distribution; a dedicated flight-deck annunciator for control-unit status	[GIVEN]
HMS	Continuous recording from both data buses on two independent recorders; a turnaround readiness report compiled at shutdown and reviewed by ground crew before release	[GIVEN]

VAY-SDD-0001 §4 describes six routings — R-1 through R-6 — carrying high-voltage power, low-voltage power, coolant and the two data buses across the aircraft's zone boundaries, several of them explicitly on opposite sides of the fuselage or through physically separated paths through the wing carry-through. That segregation is a structural and electrical design property; it is named here because several of the crew-facing facts in the table above rest on it without saying so outright. The two displays' separate graphics processors and separate low-voltage feed segments (§6), the two data buses each display can draw a source from, and DAA's two independently cross-checked processing channels are each an expression, at the flight-deck end, of the same left/right and bus-A/bus-B split that the routings maintain along their whole length. A crew-relevant conclusion that depends on that segregation continuing to hold all the way to the panel the pilot reads is exactly the kind of conclusion this document leaves to the analyses that follow it.

11 What this document does not do

This document is written the way a flight-deck design description is written, and it stops where such a document stops.

It tells you what the pilot is given to look at, what they can touch, what the aircraft will tell them and by what means, and under what conditions they will be doing it, alone, for up to fourteen legs a day. It does **not** tell you:

- which functions are allocated to the pilot, to automation, or to both;
- what tasks the pilot performs, in what order, in what time, at what workload;
- what the pilot can get wrong, how it would be detected, or how it would be recovered;
- which condition produces a Warning, a Caution or an Advisory;
- what the pilot must know at each stage of flight, or whether the flight deck lets them know it;
- whether any of the above is adequate, compliant, or acceptable;
- what the minimum flight crew is.

That omission is deliberate and it is not a gap. Deriving those things is the work. Reading that page selection between the two displays is manual while their behaviour on a processor loss is unstated implies something about reversion awareness in a cockpit with nobody else to notice a stuck page; that a recovery-system annunciator is described as presented through the display system rather than through an independent path is a claim about failure independence that has to be tested rather than assumed; that restraint status is recorded “where sensed” is a claim about detection coverage across six seats that a pilot working alone cannot verify by cross-check; that an aural alert tone which shares its path with the radio and the intercom has no stated arbitration order cannot be relied on by any analysis that assumes the single pilot hears the right thing first, every time, with nobody in the other seat to catch what they miss.

Nor does this document say anything about the passenger side of the aircraft beyond the flight-deck controls that touch it directly — door latches, cabin lighting, environmental outlets, and the door and restraint indications listed in §7.3 and §8.4. What the six passengers experience during boarding, flight and disembarkation, and what if anything they are told or shown, is outside this document’s scope entirely; it covers the flight deck, not the cabin.

11.1 Given, preliminary, assumed

Given — from the specification or VAY-SDD-0001: one pilot with no cabin crew, seated on the centreline; six passengers in the cabin; the flight deck open to the cabin above seat height; boarding as a powered, occupied ground operation with the pilot remaining seated; two reconfigurable display surfaces plus a standby display, with page selection manual and either surface able to show either page; a single centre inceptor and a single energy lever, each with two independent signal paths to the two flight control computers; no autopilot, autothrottle or flight director; continuous fly-by-wire control laws for hover, wing-borne flight and transition, the transition itself sequenced automatically; a guarded recovery handle and a separate safe-and-arm switch, both required to fire, with a dedicated firing battery and a dedicated annunciator; cabin door and latch status shown on a dedicated page and consulted at boarding and before departure; restraint status recorded “where sensed”; a turnaround readiness report compiled at shutdown and reviewed by ground crew before release.

Preliminary — expected to move: panel-zone allocation; display format assignment; phase durations.

Assumed — everything else in this document, including the accommodation range, the training-route description, and the alerting philosophy. Each should be recorded as an assumption when it is picked up, and challenged.

11.2 The open items

Numbered through §4 to §10 and repeated here for tracking: 1–6 geometry and access; 7–10 displays; 11–14 control actuation; 15–19 automation; 20–25 environment; 26–29 procedures. Two further items are flagged in place rather than numbered: the independence of the emergency-recovery annunciator from display and avionics failure modes (§8.3), and the pilot’s part, if any, in the turnaround structural inspection (§10.3, STR row).

Any human-factors conclusion that depends on one of these is provisional on it, and should say so.

Appendix A — Normal operating procedures

A.1 What these procedures are, and what they are not

Neither VAY-SDD-0001 nor VAY-HF-0001 §1–§11 contains a procedure. Between them they describe an architecture — sixteen systems, a display suite, a control set, an alerting capability — and say nothing about the order in which a pilot uses any of it. Appendix A is therefore **constructed from that architecture**, written out so that the task and error analyses this document exists to feed have something concrete to work from instead of inferring a sequence from a system table every time one is needed.

That has the same three consequences it would have for any aircraft, stated here because they should be read before any step below is used.

This is not an approved AFM, POH, or operator procedure. No flight test has been flown, no simulator campaign has been run (VAY-HF-0001 §3.1 records that a full-flight simulator is a planned asset, not an available one), and no operator has reviewed a single line of it. Status throughout is [ASSUMED] unless a step is marked otherwise, and a step marked [GIVEN] is given only as to the *system behaviour* it rests on — never as to the crew action, the wording, or the sequence built around that behaviour.

Speeds, times, quantities and masses are deliberately absent. No duration of any kind — how long a phase runs, how long a step takes, how long a margin lasts — appears anywhere below; VAY-HF-0001 §2.2 carries the phase durations that exist, and this appendix points to that section by reference rather than repeating a figure from it. Where a real procedure would give an approach speed, a rotor RPM, an energy threshold, or a landing distance, this one says [TBD].

No task analysis has been performed on any of it. These procedures state what the pilot does. They do not decompose that into how the task divides, which step could be got wrong or how, what it costs the pilot to do it, or whether the aircraft asks one person for more than one person can give. That is the work the analyses do; this appendix is their input, not a pre-emption of their conclusion.

One consequence is specific to this aircraft and worth stating before §A.2 develops it. Every procedure below is written for a flight deck with **one** occupant. There is no second pilot to read a step back, confirm a setting, or notice what the first pilot missed. A step that reads as obviously fragile because nobody is there to catch an error in it is a finding worth raising against the design, not a gap in the writing of the procedure.

A.2 Crew complement and the absence of a duty split [GIVEN]

One pilot, on the aircraft centreline. Six passengers, aft, in the cabin. No cabin crew. VAY-HF-0001 §3 states plainly that the operating convention is a “single continuous pilot duty” and that “no Pilot Flying / Pilot Monitoring exchange is possible” — there is one seat, so there is no second occupant to exchange the role with in the first place.

That is stated here as the governing fact for everything that follows, not as a finding about whether it is enough. VAY-HF-0001 §3 is explicit that the one-pilot figure “is an input, not an answer,” and this appendix inherits that position without modification. Every procedure below is written for a crew of one because that is what the specification gives; whether it should be is the minimum-flight-crew determination, and this document does not make it.

Three things follow directly from having one seat, and each is worth stating plainly because a two-crew procedure would not need to say it.

Every check in this appendix is the pilot’s own. Where a two-crew procedure would have one pilot set something and the other confirm it, this aircraft has one pilot do both — read the indication, make the selection, and re-read the indication, in sequence, because there is nobody positioned to do the second half of that pair independently. No step below is written as a challenge and a response, because there is only one voice on the flight deck to challenge or respond.

Ground staff are a separate party, not a second crew member. VAY-HF-0001 §4.1 and §7.3 both record that passenger-readiness confirmation before departure falls to “the pilot or ground staff” — without settling which, or how the two would divide it if both are involved. Where a step below involves ground staff, it is written as a handover between two parties, not as a division of one crew’s duty, and the cross-check made at that handover is [TBD] — open item 29 in the body document, carried into every relevant step below.

There is no jumpseat and no way to observe the pilot flying the aircraft as flown. VAY-HF-0001 §3 records that no jumpseat or observer position is described on the flight deck, so a check pilot or instructor cannot occupy it

during a revenue sector. Nothing in this appendix assumes a second person is ever present for training, checking, or handover purposes; where a step would benefit from one, that absence is named rather than worked around.

A.3 Checklist medium and philosophy [ASSUMED]

Neither document describes a checklist. VAY-HF-0001 §10.1 already carries this as open item 26 — “backup medium for procedures — paper, electronic, or none, is not defined” — and nothing found while writing this appendix narrows it. What follows is the minimum that can honestly be assumed, not a design.

The alerting capability at §8.1 of the body document gives an alert message, an aural tone, and for some conditions a light. Acknowledgement is a glareshield switch, after which the pilot can call up the associated system page — but whether that page carries the alert’s procedure or only the system’s status is [TBD], open item 28. A normal checklist, if one exists in electronic form, most plausibly rides the same two display surfaces that carry everything else; nothing in VAY-SDD-0001 confirms that a normal-checklist page exists at all, as distinct from the synoptic and status pages §6.2 lists. Whether normal procedures are presented electronically, on paper, from memory, or in some combination is [TBD] — open item 26 again.

Challenge-and-response has no meaning here. With one voice on the flight deck, a normal checklist can be read and actioned by the same person, or flown without being read at all and merely referenced afterward — the two are procedurally different and this document does not choose between them, because nothing in either source document does. The steps below are written as a sequence of pilot actions and confirmations without a stated medium, consistent with the honest gap this section records rather than papering over it with an assumed challenge-and-response convention borrowed from a two-crew aircraft it would not fit.

No default power-up state for any panel is described in VAY-SDD-0001 — no overhead panel is confirmed to exist at all (VAY-HF-0001 §4.2), so a “dark-cockpit” convention cannot honestly be assumed for it the way it might be for a conventional layout. The steps in §A.4.1 therefore state positively what is confirmed, rather than relying on an assumed starting condition for anything not explicitly checked.

A.4 Ground procedures

A.4.1 Flight deck preparation and power-up

1. Ground or network power established, or the aircraft’s own stored energy used directly — VAY-SDD-0001 describes no auxiliary power unit or third source distinct from the two battery packs and whatever ground supply a pad offers [GIVEN]. Which source is normal for flight-deck power-up before departure, and how the transfer between them is confirmed, is [TBD].
2. Displays — two reconfigurable surfaces, each on its own graphics processor, plus the standby display [GIVEN]. Confirm both main surfaces present a picture and that each can be set, at the bezel keys, to any available page. Confirm the standby display presents attitude, airspeed and altitude equivalents; its independence from the primary displays’ supply, sensor and processing paths is not stated [TBD] — open item 10.
3. Flight-path and energy inputs — centre inceptor and left-hand energy lever. Each sends two independent signals to the two flight control computers [GIVEN]. Confirm both respond and that no degraded-law indication is presented; whether any mode or law annunciation exists at all to confirm against is [TBD] — open item 15.
4. Energy state — HVB charge and health for both battery packs, checked against the figures displayed continuously on the power/energy page [GIVEN]. Confirm both packs report and that no fault is shown.
5. Electrical distribution — EPD bus voltage, tie-contactor state and isolation resistance for both high-voltage bus sections, displayed on the same page [GIVEN]. Confirm nominal on both sections.
6. Thermal — TMS coolant temperature on each of the two loops, and battery and inverter temperatures, displayed alongside the electrical data [GIVEN]. Pump speed on each loop is modulated automatically from reported heat load and takes no crew selection [GIVEN]; confirm temperatures only.
7. Emergency recovery system — safe-and-arm switch position confirmed against the phase of flight (ground, in this case), and the dedicated annunciator near the activation handle checked for armed state and fault [GIVEN]. Whether that annunciator’s supply and processing are independent of the two primary displays’ own failure modes is [TBD] — flagged at §8.3 of the body document and not resolved by this appendix.
8. Crew oxygen — **not applicable.** VAY-SDD-0001 describes no oxygen system, and VAY-HF-0001 §3 records the absence as a design fact rather than an oversight: cruise altitude is well below where crew oxygen would ordinarily be a design feature. No check exists because nothing exists to check.

9. Navigation reference — one inertial measurement unit and one satellite positioning receiver, both singular rather than duplicated [GIVEN]; forward air data probes; low-speed sensors reporting independently at each boom root. Confirm each source valid. What the primary displays do if the single inertial or satellite source is lost is [TBD], noted at §6.1 of the body document without being carried into its own numbered open item there; it is nonetheless a real gap in this step and is picked up at open item 32 below.
10. Alerting list — confirm no open alert on either display, or account for each individually. With one reader and no second set of eyes, an alert accepted here and forgotten has nobody else positioned to notice it later.
11. Flight management — route loaded. Loading plan and passenger data, where they affect mass and centre of gravity, are addressed at §A.4.3.

A.4.2 Exterior inspection [ASSUMED]

Neither document describes an exterior inspection, who performs it, or what it covers. VAY-HF-0001 §10.3 records only that ground crew inspect gear-bay, battery-bay and boom access panels **during turnaround**, with the pilot's part in that inspection, if any, left unstated. Whether a first-flight-of-the-day walkaround exists as a distinct, pilot-performed step — separate from the ground crew's turnaround inspection — is [TBD], open item 39. What follows is written on the assumption that some exterior check precedes the day's first departure, because it is difficult to imagine the type flying without one; its content below is constructed, not sourced.

The aircraft has no hydraulic system in its crew-facing description — flap, gear and flight-control actuation are electromechanical throughout (VAY-HF-0001 §7.1, ACT and LGS rows of §10.3) — so this inspection has none of the fluid, quantity or leak checks a hydraulically-actuated type would carry. Four items are particular to this aircraft:

The eight lift rotors, on their two booms, each with its own stop-and-align mechanism [GIVEN]. In the cruise-stowed condition the blades lock edge-on to the airflow; before the day's first vertical departure they are inspected in whatever position the stop-and-align mechanism leaves them, for blade condition and for freedom of the mechanism itself. No indication is described for confirming that condition from outside the aircraft, as distinct from the flight-deck indication checked in §A.4.1; the walkaround is a visual check standing in for one.

The two wing-mounted cruise propellers, inspected for blade condition in their stowed orientation, on the same basis as the lift rotors above.

The landing gear — a retractable, wheeled, four-point arrangement, with a mechanical downlock at each leg independent of continued actuator power [GIVEN]. Inspected for general condition and for whatever downlock indication, if any, is visible from outside the aircraft; none is named in either document.

The two cabin doors, each with an independent electric latch actuator and its own manual release lever [GIVEN]. The manual release lever at each door is checked for accessibility and condition — it is the only non-electric means of opening a door that either document describes, and its value depends on it working when the electric latch does not.

A.4.3 Energy state and dispatch decision [GIVEN] as to the constraint, [ASSUMED] as to the procedure

The operating concept gives energy replenishment only at defined points in the network, not at every pad (VAY-HF-0001 §2, [GIVEN]). That is the direct analogue of a fuel-availability constraint on a conventional type, and it produces the same shape of dispatch question — can the airframe, as it stands, complete the sector or sectors ahead of it and still reach a point where it can recharge — without either document supplying an answer.

1. HVB charge and health for both packs, and TMS coolant and cell temperatures, checked against the day's planned sequence of legs. Both packs' state feeds FCS "for energy planning" [GIVEN]; what that planning function shows the pilot, if anything beyond the raw charge and temperature figures, is [TBD].
2. Confirm the pad or pads ahead on the planned sequence include a network charging point within whatever margin the operator holds, or that the day's remaining legs can be flown and closed out on-board charge alone. **No minimum-energy threshold, display cue, or decision point is named in either source document** — this is stated as a design fact at VAY-HF-0001 §9 without being given its own tracking number there; it is carried forward here as open item 31, because a dispatch decision cannot honestly be written as a checklist item against a threshold that does not exist.
3. Where a pad ahead is a charging point, confirm (or plan to confirm on arrival) that ground charging equipment is available and compatible. EPD reconfigures contactors to route charge current "under FCS supervision," and each pack's controller accepts current independently [GIVEN]; whether that supervision needs a pilot-initiated step or runs without one is [TBD] — open item 30, picked up again at §A.9.

4. Dispatch fitness of the airframe otherwise — no minimum equipment list exists for this type in either document, and the procedural form of a dispatch decision (who makes it, against what, and where it is recorded) is [TBD], in the same shape as the energy question above but without even a partial answer. Unlike a type with line maintenance available at intermediate stops, VY-6 has none [GIVEN], so a deferral accepted at the start of a duty day has to hold for every one of the eight to fourteen sectors that follow it, with no maintenance organisation and no second pilot anywhere in that day to revisit the decision.

A.5 Passenger handling on the pad

A.5.1 Boarding with the aircraft powered [GIVEN] as to the condition, [ASSUMED] as to the sequence

Boarding is a ground operation conducted with the aircraft powered and the pilot remaining in the seat throughout [GIVEN]. It is the normal condition for every one of the type's eight to fourteen daily turnarounds, not an exceptional event, and it happens with nothing separating the flight deck from the cabin but open space above seat height — VAY-SDD-0001 describes no door or partition [GIVEN].

1. Doors — release commanded once the aircraft is confirmed at the pad [GIVEN]. The flight-deck control by which that release is actually commanded is not named — a dedicated panel, a function on an existing one, or something else — [TBD].
2. Cabin lighting and environmental outlets — set for boarding. Both are adjustable by the pilot from the flight deck [GIVEN].
3. Passenger-address announcement — sequenced automatically from flight-phase information as boarding begins, with no pilot input required [GIVEN]. The pilot can override from the flight-deck microphone [GIVEN]; content of any manual override is not addressed by either document.
4. Passenger boarding itself — conducted by the passengers, by ground staff, or by some combination; neither document names who assists whom physically, only that no cabin crew does it [GIVEN].
5. Readiness confirmation — the pilot or ground staff confirm each passenger is seated and restrained before doors are requested closed [GIVEN]. How that confirmation is made is [TBD] — the pilot's direct sightline over the open flight-deck/cabin boundary, a display indication, a ground-staff signal, or more than one of these are all named as possibilities in VAY-HF-0001 §4.1 and §7.3 without any being selected. This is open item 29 from the body document, and it is the single largest procedural hole in this section: every other step here can be written with some confidence about what the pilot does; this one cannot, because the method of confirmation is not settled.

A.5.2 Restraint and door closure [GIVEN] as to the equipment, [ASSUMED] as to the sequence

1. Restraint status — six four-point restraints, two-two-two, checked against the cabin synoptic page [GIVEN]. Buckle status is recorded “where sensed,” a qualifier VAY-HF-0001 §7.3 and §8.4 both use without stating which of the six positions are instrumented and which are not. For any position not reporting, confirmation is by whatever means §A.5.1 step 5 above resolves to, once it is resolved; today it is [TBD].
2. Door and latch status — both doors confirmed closed and latched on the cabin synoptic page, described specifically as consulted “during boarding and before departure” [GIVEN]. This is one of the two conditions VAY-HF-0001 §8.4 singles out by name for crew attention at these points in the turnaround; the other is restraint status above.
3. Manual release levers — confirmed stowed and not fouled, at both doors.
4. Cabin lighting and environmental outlets — reset from the boarding configuration if different for departure.

A.5.3 Passenger briefing [TBD] throughout

Neither document addresses passenger briefing content or medium. What is established is the boundary of what the flight deck can do: an automatic passenger-address announcement sequenced by flight phase, and a pilot override from the flight-deck microphone [GIVEN], §A.5.1 above. Nothing beyond that — restraint-use instruction, brace guidance, emergency egress information, or anything said about the guarded emergency-recovery system at the centre console — is named in either document. VAY-HF-0001 §11 states explicitly that the passenger side of the aircraft is outside its scope except at the flight-deck controls that touch it directly; this appendix inherits that boundary rather than filling it in. Passenger briefing content and delivery medium is open item 37.

With no cabin crew to deliver a live briefing from within the cabin, and the pilot seated forward through the whole of boarding, whatever briefing passengers receive is either recorded content carried on the automatic announcement,

a safety card or equivalent placed in the cabin, the pilot’s microphone override, or some combination — none of which either document names.

A.6 Departure

A.6.1 Before start

1. Restraint and door closure — confirmed complete (§A.5.2).
2. Energy state and dispatch margin — confirmed (§A.4.3).
3. Mass and centre of gravity — six passengers of varying individual mass, loaded leg to leg with a different combination each time, change the aircraft’s mass and inertia distribution in a way VAY-HF-0001 §9 records as unresolved for the control-law and transition-threshold schedule (open item 19). Whether a load-and-balance computation is performed before each departure, by what means, and against what limits, is not stated by either document [TBD] — open item 38.
4. Departure briefing — including the loss of one propulsor. Ten independent inverter-motor units, one per propulsor, with the flight control system redistributing torque across the survivors on the loss of any one, and no gearbox coupling a failure to a neighbour [GIVEN]. No described crew action follows a single propulsor loss during a vertical departure — the redistribution is stated as automatic and nothing in either document names a pilot response to confirm, refuse, or act on top of it. The briefing states what is known: that the system redistributes automatically. What the pilot would additionally do is [TBD].
5. Take-off, or vertical-departure, configuration check — **no such check is described in any form.** VAY-HF-0001 §10.1 states this plainly: no configuration monitor, and no stated parameter set for one, appears anywhere across the sixteen system descriptions. Where a conventional transport would have a discrete configuration-monitor input covering flaps, trim, doors and the like, this aircraft has nothing named at all. Whether some check exists that has simply not been described, or whether none exists, is [TBD] — open item 32. The steps at §A.6.2 below are the closest this appendix can honestly come to one, built from the individual indications that do exist rather than from a monitor that does not.

A.6.2 Rotor spin-up and pad checks

1. Lift rotors — eight, on two booms, released from the cruise-stowed edge-on position by the stop-and-align mechanisms and brought to a condition ready for hover thrust [GIVEN] as to the mechanism, [TBD] as to what confirms it from the flight deck. No indication is named for distinguishing “rotors ready” from “rotors stowed” on either display.
2. Weight-on-wheels — sensed at each main gear leg, driving the flight control system’s automatic selection between ground and air behaviour with no pilot selection [GIVEN].
3. Traffic and terrain — dedicated DAA page called up, described in VAY-HF-0001 §6.2 as “monitored during pad approach and departure” [GIVEN]. Confirm the picture presented and no unresolved track close to the pad.
4. Restraint and door/latch status — final confirmation, repeating §A.5.2 steps 1 and 2. These two are checked again here because they are the two conditions VAY-HF-0001 §8.4 names specifically as consulted “before departure,” distinct from being checked once during boarding and assumed to hold.
5. Energy, electrical and thermal indications — confirmed nominal, per §A.4.1 steps 4 through 6, immediately before commanding a departure.

A.6.3 Vertical departure

1. Centre inceptor — commands flight path in the hover control law; energy lever commands power. Both send independent signals to the two flight control computers [GIVEN]. What envelope protection, if any, is provided in hover is [TBD] — open item 16.
2. Displays — the pilot selects, at the bezel keys, whichever pages the departure needs across the two re-configurable surfaces: a flight page, the traffic and terrain page, and whatever else the moment calls for. Page selection is manual throughout, with no stated default assignment by phase [GIVEN] as to the manual selection, [TBD] — open item 8 — as to any default. With one reader and two surfaces to divide between a flight page and everything else, what is not on either surface at a given moment is, for that moment, not being read.
3. Climb continues to whatever height precedes the transition; VAY-HF-0001 §2.2 treats the vertical-departure phase and the transition below as one figure rather than describing a separate climb segment ahead of it.

A.6.4 Transition to wing-borne flight

The transition is sequenced by the flight control system, not commanded by a discrete pilot selection. It commands the lift-rotor stop-and-align mechanisms “once conditions permit,” shifting authority from rotor thrust to aerodynamic surfaces as airspeed builds [GIVEN].

1. The pilot’s part is to continue flying the aircraft through the sequence on the same centre inceptor and energy lever used throughout the departure, and to read whatever the displays show while it happens [GIVEN] — VAY-HF-0001 §2.1 states this almost exactly: what stays with the pilot “is flying the aircraft through it and reading whatever the displays and alerting system show while it happens.” No initiating selection is made because none is described.
2. **No mode or transition annunciation is confirmed to exist.** Whether the pilot has any positive indication that the transition has begun, is proceeding normally, or has completed — as distinct from inferring it from airspeed and control feel — is [TBD], open item 15. This sequence runs, automatically, up to twenty-eight times a day for one aircraft (fourteen sectors, two transitions each, VAY-HF-0001 §2.1), flown by the same one pilot every time, and the absence of a confirmed annunciation for it is stated here as a fact worth the analyses’ attention rather than worked around with an assumed indication that would not survive scrutiny.
3. Any automatic reversion of the transition back toward hover, and any alerting associated with it, is [TBD] — open item 17. No step is written for a reverted transition because no reversion behaviour is described to write one against.

A.7 Cruise

Wing-borne cruise, per the phase split at VAY-HF-0001 §2.2, is the settled middle segment of the sector, flown at 300 to 600 m above the city on a 20 to 30 km sector [GIVEN]. Four things are read continuously, each corresponding to a system this aircraft carries that a conventional type either does not have or does not expose to the crew in the same way:

1. **Energy state.** HVB charge and health for both packs; EPD bus voltage, tie-contactor state and isolation resistance for both bus sections; TMS coolant temperature on each loop and battery and inverter temperatures — all displayed continuously [GIVEN]. As at §A.4.3, no minimum-energy threshold or decision cue exists to read this information against; the pilot has the raw figures and no stated trigger point.
2. **Traffic and terrain.** The DAA page, built from two independently cross-checked processing channels, falling back to the local sensor picture alone if the transponder path is affected, with no stated pilot action to trigger that fallback [GIVEN].
3. **Cabin state.** Door, latch and restraint status remain on the cabin synoptic page through the cruise. In flight this is indication only — neither document describes any means of adjusting a restraint from the flight deck once the aircraft is airborne, in the same way a conventional type’s crew cannot reach a cabin restraint once seated forward.
4. **Communications.** Two VHF radios and the vertiport datalink, sharing the same audio path as the alert tone generator, which is the same path the cabin intercom would use if the intercom and the flight-deck microphone connect to it [GIVEN]. Arbitration between an alert tone and a radio call on that shared path is [TBD], carried from §8.1 of the body document.

No routine crew selection is described for propulsion allocation across the ten propulsors, for thermal management, or for electrical configuration in normal cruise flight — VAY-HF-0001 §9 states each of these as automatic, with no crew-facing control named for overriding any of them in the ordinary case [GIVEN]. The routine selections that remain with the pilot are page assignment between the two displays, and cabin lighting or environmental outlet adjustment if a passenger requests it through whatever means either document leaves unstated.

A.8 Arrival

A.8.1 Descent and approach preparation

1. Destination pad assessment — an elevated structure among buildings, with shear and turbulence off nearby surfaces a normal operating condition [GIVEN], VAY-HF-0001 §10.2. **No measured wind, shear or obstacle data exists for any specific pad,** and no means is described by which the pilot learns pad-specific conditions before committing to an approach — [TBD], open item 33.
2. Landing performance — no mass or wind limit is stated for a vertical landing at a given pad, and no performance margin equivalent to a runway-distance requirement exists for this aircraft, because it does not

need a runway [GIVEN] as to the absence of the runway case, [TBD] as to any hover-landing performance figure that replaces it — open item 34.

3. Approach briefing — including the transition to hover-borne flight ahead, the building-shaded pad environment, and what the pilot intends to do if the approach cannot be continued to landing. No go-around or diversion procedure specific to a rejected pad approach is described in either document; the briefing states the intention against a procedure that does not yet exist.
4. Lighting and visual environment — day and night operation are both named [GIVEN], but pad lighting, surrounding-structure visibility at night, and low-sun conditions have no measured data behind them [TBD], open items 22 and 24 from the body document.

A.8.2 Transition to hover

The reverse of §A.6.4. As airspeed reduces, the flight control system sequences the transition back into hover, releasing the lift-rotor stop-and-align mechanisms and shifting authority from aerodynamic surfaces back to rotor thrust [GIVEN] — VAY-HF-0001 §9 states the departure sequence and adds “the reverse on the way back into hover” without describing the reverse sequence separately. The same absence of a confirmed mode or transition annunciation applies here as at §A.6.4, and it is the same open item, not a second one.

A.8.3 Approach and landing on the pad

1. The pilot flies the descent and approach on the centre inceptor and energy lever in the hover control law, reading the flight page and the DAA traffic and terrain page across the two display surfaces (manual selection, §A.6.3 step 2 applies equally here). What envelope protection exists in hover is the same open gap noted at §A.6.3 step 1.
2. External vision — over-nose and over-side cut-off angles from the design eye position are not yet defined [TBD], open item 2 of the body document, and VAY-HF-0001 §4.1 names this approach specifically as the phase where that gap matters most: close-quarters visibility of the pad edge and the surrounding structures depends on it directly.
3. Landing gear — extension is pilot-commanded, not automatic; each leg is observed individually on the display as it cycles, with a manual extension handle available if the electric actuators do not respond [GIVEN]. Confirm all four legs down, with the mechanical downlock at each leg independent of continued actuator power [GIVEN].
4. Touchdown — weight-on-wheels sensed at the main legs, and the flight control system’s ground/air behaviour selected automatically with no pilot action [GIVEN]. No wind limit, mass limit, or flare technique is stated for the landing itself [TBD], the same gap noted at §A.8.1 step 2.
5. Ground manoeuvring on the pad, if any is needed to position the aircraft after touchdown — neither document describes rudder pedals, a yaw pedal, or any directional-control input separate from the centre inceptor [TBD], open item 14 of the body document. Whether the aircraft repositions itself on the pad surface at all after a vertical landing, and by what control input if so, is accordingly unresolved; no taxi procedure is written here because nothing in either source document describes a taxi capability to write one against.

A.9 Turnaround

The turnaround is a powered, occupied event, not a shutdown. The aircraft stays energised, the pilot stays in the seat, and this repeats eight to fourteen times a day [GIVEN], VAY-HF-0001 §2.1 (§2.2 gives the phase’s nominal length). Nothing below assumes the systems secured at §A.10 are touched during a normal turnaround; that section describes a different, end-of-day event.

1. Deboarding — the reverse of §A.5, with the arrival passenger-address announcement sequenced automatically from flight-phase information and the same pilot microphone override available [GIVEN].
2. Door release — commanded once the aircraft is confirmed at the pad [GIVEN], by whatever flight-deck control §A.5.1 step 1 eventually resolves to.
3. Ground charging, where the pad is a network charging point — EPD reconfigures contactors “under FCS supervision” [GIVEN]; whether that requires a pilot-initiated step, runs automatically once ground equipment connects, or something between the two, is [TBD], open item
30. Where the pad is not a charging point, no replenishment occurs and this step does not apply — the day’s energy plan from §A.4.3 has to already account for that.
4. Next-leg boarding, restraint and door closure — the whole of §A.5, repeated. On a short turnaround with no cabin crew, this is the pilot’s task alongside whatever charging monitoring step 3 above adds, with nobody

else on the flight deck to divide the window between them.

5. Turnaround readiness — VAY-HF-0001 §10.1 records that a turnaround readiness report is compiled by HMS **at engine shutdown** and reviewed by ground crew before the aircraft is released for its next flight [GIVEN]. A normal turnaround, by the design recorded at §2 of the body document, does not shut the aircraft down. Whether any check exists at each of the eight to fourteen daily turnarounds that do **not** trigger that report — as distinct from the report generated once, at the end of the duty day — is not stated by either document [TBD], open item 35.

With no line maintenance available at the pad [GIVEN], any anomaly noticed during a turnaround has no local remedy. The dispatch judgment made once at §A.4.3, before the first sector of the day, is in effect asked to hold across every turnaround that follows it, without a maintenance organisation and without a second pilot anywhere in the day to revisit it. That is stated here as a consequence of the operating concept, not as a finding that the concept is unworkable.

A.10 Shutdown and handover

Unlike the turnaround at §A.9, this section describes the end of a duty period or a return to a network point for a full stop — not a normal between-sector event.

1. Systems secured — HVB packs, EPD bus sections and contactors, TMS pumps, brought to their shutdown state. No sequence is stated in either document [TBD].
2. Emergency recovery system — safe-and-arm switch set to its ground state, consistent with the phase-of-flight logic described at VAY-HF-0001 §9 [GIVEN] as to the switch existing and being phase-dependent, [ASSUMED] as to this specific selection.
3. Turnaround readiness report — compiled by HMS at shutdown, from continuous recording on both data buses across two independent recorders [GIVEN]. This is the report referred to but not generated at §A.9 step 5 above; here, at an actual shutdown, it is produced.
4. Handover to ground crew — the report is reviewed at a maintenance terminal before the aircraft is released for its next flight [GIVEN]. The pilot's own part in that release decision, beyond calling up recent recorded events on a maintenance page if needed, is not stated [TBD].
5. Handover to another pilot, where the aircraft's next duty is flown by someone else — content and method are [TBD], open item 36. With no jumpseat or observer position on the flight deck (VAY-HF-0001 §3), an in-person handover briefing conducted on the aircraft, in the way a two-crew type might manage one, has no seat to happen from.

A.11 Normal-procedure open items

These are the holes in Appendix A specifically, numbered onward from the twenty-nine items already tracked at §10.1 and §11.2 of the body document.

30. **Ground/network charging initiation.** Whether the pilot-facing side of EPD's charge-contactor reconfiguration is a pilot-initiated step, an automatic one, or something between the two (§A.4.3, §A.9).
31. **Energy dispatch threshold.** No minimum-energy figure, display cue, or decision point for completing a sector and reaching a network charging point is named anywhere in either document (§A.4.3, §A.7).
32. **Pre-departure configuration check.** Whether anything resembling a take-off or vertical-departure configuration check exists, and what it covers if so — none is described in any form in VAY-SDD-0001 (§A.6.1, §A.6.2).
33. **Pad-specific condition assessment.** How the pilot learns wind, shear, obstacle and lighting conditions at a specific destination pad before committing to an approach (§A.8.1).
34. **Vertical-landing performance data.** Mass and wind limits for a landing at a given pad; no figure of any kind replaces the runway-distance requirement this type does not need (§A.8.1, §A.8.3).
35. **Turnaround-level readiness check.** Whether anything checks the aircraft at each of the eight to fourteen daily turnarounds that do not trigger the shutdown-generated HMS readiness report (§A.9).
36. **End-of-duty handover.** Content and method of a handover between pilots at change of duty, with no jumpseat or observer position to support one delivered on the aircraft (§A.10).
37. **Passenger safety-briefing content and medium.** Beyond the automatic phase-triggered announcement and the pilot's microphone override, nothing is named for restraint use, brace guidance, egress, or the emergency-recovery system (§A.5.3).
38. **Load and centre-of-gravity computation.** Whether, and how, a load-and-balance check is performed before each departure given that six passengers of varying mass change the loaded condition leg to leg (§A.6.1).

39. **Pre-flight exterior inspection.** Whether a pilot-performed walkaround exists ahead of the day’s first departure, distinct from the ground crew’s turnaround inspection of gear-bay, battery-bay and boom access panels, and what it would cover if so (§A.4.2).

Any procedure step above that rests on one of these — and most do — is provisional on it, in exactly the sense §11.2 of the body document already states for items 1 through 29.

Appendix B — Non-normal procedures

B.1 What these procedures are

Neither VAY-SDD-0001 nor VAY-HF-0001 §1–§11 contains a procedure, and Appendix A already states plainly that this document constructs procedures from the architecture rather than reporting ones that already exist. The same is true here, with one further complication Appendix A did not carry: a non-normal condition is, by definition, one the architecture was not necessarily built to make easy to read, and constructing a procedure for it means saying, honestly, when the architecture gives the pilot nothing to read at all.

Everything below inherits Appendix A’s governing fact without repeating its full statement: one pilot, on the centreline, with no second occupant to confirm a reading, repeat a step back, or notice what the first reading missed. Where Appendix A could note that absence once at §A.2 and let each later step stand on it, a non-normal procedure needs the same absence stated at the point it bites hardest — where a two-crew flight deck would routinely divide a look at the alert list from a hand on the aircraft, and this one cannot.

Each entry below describes an indication and an action: what appears on a display, an annunciator or a light, and what the pilot does about it, once. Where the source material gives no indication, or no action, or neither, the entry says so rather than inventing one. Conditions with no stated action of any kind are marked in the text and collected at §B.15 — the intended purpose of that section is to give the safety and human-factors assessments a starting list of the places a design decision has to close a gap that no procedure can, not to catalogue the design’s shortcomings for their own sake.

No memory-item set is proposed here, for the same reason Appendix A does not propose a checklist medium: VAY-HF-0001 §10.1 carries “memory (recall) item set” as open item 27, unresolved, and nothing in this appendix’s construction narrows it. Several entries below read as needing to be done before any list could plausibly be read; each is left exactly that unresolved for the analysis to take up, not pre-empted with a memory-item designation this document has no basis to assign.

B.2 Battery and energy (HVB)

Two independent battery packs, each of four strings, feed the aircraft’s two high-voltage bus sides through a normally-open tie contactor. The packs sit in a sealed bay beneath the cabin floor, vented overboard. Charge and health data for both packs are displayed to the pilot continuously, on the same power/energy page that carries EPD’s bus voltage, contactor state and isolation resistance, and TMS’s coolant and cell temperatures (VAY-HF-0001 §6.2).
[GIVEN]

B.2.1 Single string isolated

1. Indication — the power/energy page shows whatever change a string’s isolation produces in the affected pack’s reported charge or health figure. Whether the page carries a status distinct at the level of an individual string, as opposed to the pack-level charge and health VAY-HF-0001 §6.2 names, is not stated by either source document; nothing in the architecture as described distinguishes “one string down” from “the pack’s aggregate figure moved.” [TBD] — open item 40.
2. Action — confirm the affected pack’s remaining charge and health against the day’s planned sequence of legs (§A.4.3 of Appendix A applies equally in flight). No contactor selection is named for isolating or restoring a single string from the flight deck.
3. Continue the sector, or revise the day’s remaining legs against the reduced pack capacity, using the same raw charge and health figures Appendix A already notes have no stated minimum-energy threshold behind them (open item 31).

B.2.2 Pack temperature rise

1. Indication — battery and inverter temperatures for both packs, displayed continuously alongside per-loop coolant temperature on the power/energy page. [GIVEN]
2. Action — coolant pump speed on each loop is modulated automatically from reported heat load, with no described crew selection (VAY-HF-0001 §9). §7.2 of the body document names exactly this gap — “selecting a coolant-loop configuration” — as a thing the systems do without a stated flight-deck control for a pilot override. [TBD] — open item 41.
3. The only action either document supports is a change to the flight itself — reducing commanded power through the energy lever, or ending the sector — rather than any direct intervention in the thermal system.

B.2.3 Energy below plan

1. Indication — the same continuous charge, health, bus and coolant figures read throughout normal cruise (Appendix A §A.7 item 1), with no minimum-energy threshold, display cue or decision point named anywhere in either source document. [TBD] — open item 31, restated here because it applies to an in-flight recognition of the condition, not only to the pre-departure dispatch decision Appendix A tracks it against.
2. Action — assess the figures against the day’s planned sequence of legs and the location of the nearest network charging point (§A.4.3), and revise the plan — divert to a charging point, or curtail the remaining legs — using judgement the design gives no cue to anchor.

B.3 Electrical distribution (EPD)

Two high-voltage bus sides, joined by a single normally-open tie contactor; bus voltage, contactor state and isolation resistance for both sides displayed continuously. [GIVEN]

B.3.1 Loss of one high-voltage bus side

1. Indication — bus voltage and contactor state on the power/energy page show the affected side unpowered.
2. Action — whether the tie contactor can be closed by a pilot selection to feed the affected side’s loads from the remaining side, whether closure is automatic, or whether it is not available at all in flight, is not stated. §7.2 of the body document names “closing or opening the electrical bus-tie contactor” specifically, without a stated flight-deck control for pilot override or manual selection. [TBD] — open item 42.
3. Confirm which propulsors, displays or other loads draw from the affected side — §10.3’s routing note records that the two displays’ separate graphics processors and feed segments follow the same left/right split as the bus sides — and continue the sector on whatever the remaining side and the automatic reconfiguration, if any, leave available.

B.3.2 Low-voltage bus loss

1. Indication — not described. VAY-SDD-0001 names a low-voltage distribution system only in passing — as what the emergency recovery system’s dedicated firing battery is independent of (§8.3 of the body document), and as the feed for cabin lighting’s two strings (§7.3) — without describing its architecture, its display, or what depends on it beyond those two mentions.
2. Action — none is stated, because neither the indication nor the consequence of a low-voltage bus loss is described anywhere in the source material. [TBD] — open item 43. This gap sits squarely inside the pattern §B.1 describes: a condition the architecture does not yet give the pilot any means of reading.

B.3.3 Insulation monitor indication

1. Indication — isolation resistance for both bus sides, displayed continuously on the power/energy page alongside bus voltage and contactor state.
2. Action — no threshold, no annunciation behaviour distinct from the continuous reading, and no procedure keyed to a falling isolation- resistance figure is named. [TBD] — open item 44.
3. In the absence of a stated procedure, monitor the reading and treat a declining trend as a reason to end the sector; nothing in either document supports a more specific action than that.

B.4 Propulsion (DEP and LFT)

Ten independent inverter-motor units — eight lift rotors on two booms, two wing-mounted cruise propellers — each without a gearbox, with the flight control system redistributing torque across the surviving units on the loss of any one. [GIVEN] Each lift rotor carries its own stop-and-align mechanism, locking its blades edge-on to the airflow for cruise. [GIVEN]

B.4.1 Loss of one lift unit in hover

1. Indication — not named. VAY-SDD-0001 states that the flight control system redistributes torque across the surviving units automatically; it does not state what, if anything, the pilot is shown to identify which of the eight lift rotors has been lost, or that any of the eight is individually annunciated at all. [TBD] — open item 45, the same shape of gap Appendix A's departure briefing (§A.6.1 step 4) already names for a single propulsor loss in general terms.
2. Action — none beyond continuing to fly the centre inceptor and energy lever through whatever redistribution the flight control system performs automatically. No crew selection is described for isolating, or attempting to restart, a lost lift unit.
3. What margin remains in hover with one of eight lift units lost — whether the aircraft continues to hover normally, hovers with reduced authority, or requires an immediate transition to wing-borne flight — is not stated by either document.

B.4.2 Loss of a lift unit group

1. Indication — as B.4.1, with the same absence of individual-unit annunciation. Whether a group loss confined to one boom is distinguished, on any display, from a single-unit loss elsewhere is not stated. [TBD] — open item 46.
2. Action — as B.4.1. Nothing in the source material describes a different crew response to a group loss than to a single loss, because neither is described with any crew response beyond continuing to fly.
3. A loss confined to one boom changes the aircraft's lift and moment distribution left to right in a way a single lost unit spread differently would not; whether the flight control system's redistribution logic, or the pilot's available control authority, treats the two cases differently is not addressed by either document.

B.4.3 Loss of a cruise unit

1. Indication — as above, not named individually.
2. Action — continue flying the wing-borne law through whatever redistribution occurs across the remaining cruise unit and, where the control law permits it, the lift rotors. No asymmetry indication distinct from handling itself is described.
3. Whether a single cruise-unit loss changes the transition threshold back to hover, given that transition timing already depends on airspeed and control-law state rather than a discrete pilot selection (§9 of the body document), is not addressed. [TBD] — open item 47.

B.4.4 Rotor fails to stop or index for cruise

1. Indication — not named. The stop-and-align mechanism's normal behaviour is described (§9, §A.4.2 of Appendix A); no indication is described for confirming, from the flight deck, that a given rotor has reached the stowed, edge-on position rather than remaining part-deployed, beyond the general absence of a rotors-ready indication already noted at Appendix A §A.6.2 step 1.
2. Action — whether the transition to wing-borne flight proceeds, is inhibited, or produces some other flight-control-system behaviour when one of eight rotors has not reached the stowed position is not stated. [TBD] — open item 48.
3. No procedure is written past that point, because no indication exists to tell the pilot which of those outcomes has occurred.

B.5 Transition

The transition between hover-borne and wing-borne flight is sequenced by the flight control system rather than commanded by a discrete pilot selection, in both directions, on every sector (§9 of the body document; §A.6.4 and §A.8.2 of Appendix A). There is no tilting mechanism; the change is a matter of control law and airspeed acting on the same fixed lift rotors and cruise propellers throughout. [GIVEN] No mode or transition annunciation is

confirmed to exist in either direction — open item 15 — and this section inherits that absence rather than repeating it at every step below.

B.5.1 Transition to wing-borne flight does not complete

1. Indication — none beyond what the pilot reads directly from airspeed, attitude and control feel, in the same absence of annunciation already recorded at Appendix A §A.6.4 step 2.
2. Action — no crew-initiated means of aborting or repeating the transition sequence is described. Whether the flight control system reverts automatically toward hover if the sequence does not complete, and what that reversion looks like to the pilot if it happens, is [TBD] — open item 17, restated here because it is precisely the gap this entry needs closed. The absence of any crew-initiated abort, distinct from an automatic reversion, is [TBD] — open item 49.
3. The pilot's available action is limited to continuing to fly the centre inceptor and energy lever through whatever the flight control system does next, because no discrete crew procedure exists for a transition that does not complete on its own.

B.5.2 Transition to hover does not complete

1. Indication — as B.5.1, in the reverse direction, on approach to a pad rather than departing one.
2. Action — as B.5.1; the absence is the same gap, not a second one, and it carries a particular weight on this side of the sector. A transition to hover that does not complete happens during the approach and landing phase, over an elevated pad among buildings, with the external-vision gap already named at Appendix A §A.8.3 step 2 compounding whatever the pilot can see of the aircraft's own state.
3. No go-around or waved-off approach procedure specific to an incomplete transition is described, beyond the general absence of any rejected-approach procedure already noted at Appendix A §A.8.1 step 3.

B.6 Thermal management (TMS)

Two independent coolant loops, left and right, with pump speed on each modulated automatically from reported heat load and no described crew selection. [GIVEN]

B.6.1 Loss of one coolant loop

1. Indication — coolant temperature for the affected loop, together with the battery and inverter temperatures it serves, shown on the power/energy page moving outside its normal reading.
2. Action — as at §B.2.2, no coolant-loop reconfiguration control is named for the pilot (open item 41). The available response is the same reduction in commanded power through the energy lever, or an earlier end to the sector, rather than any direct system intervention.
3. Which battery pack and which propulsor group each loop serves, and therefore what the practical consequence of losing one loop is for the equipment on the other side of the aircraft, is not stated by either document.

B.6.2 Loss of both loops

1. Indication — as B.6.1, on both sides at once.
2. Action — as B.6.1, with no reconfiguration available on either loop and no described means of shedding thermal load beyond reducing commanded power.
3. What condition the battery packs, inverters or motors reach, unmanaged, over the remainder of a sector with no coolant flow is not characterised in either document, and no figure is given against which the pilot could judge how much of the flight remains available before that matters. [TBD]

B.7 Flight control (FCS and ACT)

VAY-SDD-0001 describes the flight control system as running three computing lanes, with no mechanical reversion path of any kind. [GIVEN] VAY-HF-0001 describes the centre inceptor and energy lever as each sending two independent signal paths to “the two flight control computers” (§6, §7.1 of the body document); this appendix uses the finer-grained, three-lane description because it is the one the aircraft's own specification gives, and does not attempt to reconcile the two figures here. Every command reaches the ailerons, the V-tail surfaces and the ten propulsors electrically, through electromechanical actuation, with no mechanical or hydraulic linkage anywhere between the flight deck and a control surface or propulsor. [GIVEN]

B.7.1 Loss of one computing lane

1. Indication — not described. No mode or law annunciation is confirmed to exist for any flight-control-system state (open item 15), and this is the specific case that gap most directly touches: a lane loss is exactly the kind of event a law annunciation would exist to show, and none is confirmed.
2. Action — none beyond continuing to fly, because no indication is confirmed to tell the pilot a lane has been lost in the first place. [TBD] — open item 50.
3. What handling change, if any, follows the loss of one of three lanes is not stated.

B.7.2 Loss of two lanes

1. Indication — as B.7.1.
2. Action — as B.7.1. With three lanes and no mechanical reversion, the loss of two lanes leaves one lane carrying the full command path with nothing behind it if that lane is also lost; whether that condition is annunciated any differently from a single-lane loss is not stated. [TBD] — open item 50 applies equally here.
3. This is the point at which §B.13.1 below becomes directly relevant: with no mechanical path and no described means of the pilot confirming or influencing which lane, if any, remains in command, the emergency recovery system is the design's stated means of addressing a flight-control condition beyond this one, not a procedure step within this section.

B.7.3 Actuator jam or runaway

1. Indication — not described. Actuator position and status are reported back to the flight control system and the displays generally (§10.3 of the body document, ACT row); whether a jam or a runaway is distinguished, on any display, from normal actuator behaviour is not stated.
2. Action — none is described. [TBD] — open item 51.
3. Whether the electromechanical actuation architecture provides any means of isolating a jammed or runaway actuator from the surface or propulsor it drives, as distinct from the flight control system's general redistribution of command across that surface's or propulsor's neighbours, is not addressed by either document.

B.8 Navigation and air data (NAV)

A single inertial measurement unit and a single satellite positioning receiver; forward air data probes; low-speed sensors reporting independently at each boom root. [GIVEN]

B.8.1 Loss of satellite position

1. Indication — the position source shown as invalid or unavailable on whichever display page carries it.
2. Action — what the flight displays do when the single satellite positioning source is lost is not stated by either document; VAY-HF-0001 §6.1 names the gap in passing — “whether the primary flight displays have any means of reversion if the single inertial or satellite positioning source is affected” — without giving it its own tracking number there. [TBD] — open item 52, closing that gap here.
3. With a single inertial reference also fitted, and no stated cross- reference between the two beyond both feeding the same, singular navigation function, the pilot's available action on losing satellite position is limited to whatever the inertial reference alone continues to support, and neither document says what that is.

B.8.2 Air data disagreement

1. Indication — forward air data probes are named in the plural (§10.3 of the body document, NAV row); whether their outputs are compared against each other, and what is shown if they disagree, is not stated. [TBD] — open item 53.
2. Action — none is described, because no disagreement indication is described to act on.

B.8.3 Loss of low-speed sensing

1. Indication — low-speed sensors report independently at each boom root (§10.3 of the body document); a loss at one boom would, on that description, be distinguishable from a loss at the other, though no display behaviour for either case is named.
2. Action — not described. The hover and transition control laws are stated to draw on this sensing (§9), and a loss at one boom raises the same left/right asymmetry question already noted at §B.4.2 for a lift-unit- group

loss, without either document saying whether the control law treats the two conditions the same way. [TBD]
— open item 54.

B.9 Landing gear (LGS)

A retractable, wheeled, four-point gear; weight-on-wheels sensed at each main leg; a mechanical downlock at each leg independent of continued actuator power; a manual free-fall extension handle in the flight deck, independent of the electric actuators. [GIVEN]

1. Indication — gear position is shown on the display during extension and retraction (§6.2 of the body document), with each leg observed individually as it cycles (Appendix A §A.8.3 step 3).
2. Gear does not extend normally — alternate extension by the manual handle, independent of the electric actuators. [GIVEN]
3. Downlock confirmation — each leg's downlock is mechanical and independent of continued actuator power; whether the display distinguishes a confirmed downlock from an uncertain one, on any one of the four legs individually, is not stated.
4. A partial or asymmetric downlock — one leg of four not confirmed down while the other three are — is not addressed by any procedure in either document, and the landing that would follow one, on a four-point gear rather than the multi-bogie arrangement a larger transport carries, is not characterised. [TBD] — open item 55.
5. No wind or mass limit is stated for a vertical landing in any gear configuration, normal or otherwise (Appendix A §A.8.1 step 2, open item 34), so a partial-downlock landing carries that same absence of performance data in addition to the absence of a procedure for the condition itself.

B.10 Cabin and doors (CAB)

Two cabin doors, one each side, each with an independent electric latch actuator and its own manual release lever; door and latch status presented on a dedicated cabin synoptic page. [GIVEN]

B.10.1 Door latch indication in flight

1. Indication — the cabin synoptic page carries door and latch status continuously, the same page consulted at boarding and before departure (§8.4 of the body document).
2. A disagreement or an unlatched indication appearing on that page in flight, as distinct from the pre-departure check it is named for, is not addressed by any procedure in either document. [TBD] — open item 56.
3. Action — with no in-flight door control described (only a manual release lever intended for ground use, per Appendix A §A.4.2's walkaround note), the pilot's available response to an in-flight indication of this kind is limited to whatever flight-path or landing decision follows from not being able to confirm or correct the condition directly.

B.10.2 Passenger incapacitation or interference

1. Indication — none is described. The flight deck is open to the cabin above seat height, giving the pilot a direct sightline over the passenger seats (§4.1 of the body document), and no camera or other sensor for cabin observation is described beyond that sightline.
2. Action — with no cabin crew and no means of leaving the flight-deck seat described for any in-flight purpose, the pilot's available response to a passenger condition observed over that sightline is not addressed by either document. [TBD] — open item 57.
3. Whether the sightline itself is adequate for recognising a passenger condition from the forward seat, at whatever angle and distance the eventual flight-deck geometry produces (§4.3's open geometry items), is a question this appendix cannot answer because the geometry it would depend on does not yet exist.

B.11 Pilot incapacitation

No procedure is written for this condition, because no element of the design gives one anything to be written against.

There is one pilot, on the centreline, with no second pilot seat and no jumpseat or observer position described anywhere on the flight deck (§3 of the body document). VAY-SDD-0001 describes no means by which the aircraft detects that the pilot is incapacitated, no means by which a passenger could take any flight-path or system action from the cabin, and no means by which the flight continues, is interrupted, or is landed without the one person at

the centre inceptor and energy lever remaining capable of using them. [GIVEN] as to the absence of any of these provisions in the source material; the absence itself is the finding.

This condition sits outside the indication-and-action form the rest of this appendix uses, because neither exists here. It is carried to §B.15 rather than left implicit: a document that describes what the pilot sees and does has to say plainly when there is no pilot left to see or do anything, and no other means the design provides in that circumstance. [TBD] — open item 58.

B.12 Fire, smoke and fumes

VAY-SDD-0001 describes no crew oxygen system (§3 of the body document; Appendix A §A.4.1 step 8 records this as a design fact rather than an oversight, given a cruise altitude well below where crew oxygen would ordinarily be a design feature). A smoke or fumes procedure on this aircraft accordingly has no masks-and-goggles step to reach for; whatever the pilot does about smoke or fumes is done breathing the flight deck's own air throughout.

B.12.1 Smoke or fumes on the flight deck

1. Indication — an alert message on whichever display carries it, with an aural tone and, for some conditions, a light on the alerting panel (§8.1 of the body document); no smoke detector, and no distinction between smoke originating in the flight deck itself and smoke reaching it from elsewhere, is named specifically for this condition.
2. Action — electrical load shedding and ventilation selections are the general means available; the specific sequence, and any smoke-removal procedure distinct from isolation, is not described. [TBD] — open item 59.
3. Land as soon as possible. With no oxygen equipment and no second occupant to divide attention between flying the aircraft and managing the source, this step is stated here as a description of what the flight deck offers, not as a finding about whether it is enough.

B.12.2 Battery bay indication

1. Indication — the sealed, underfloor, overboard-vented battery bay housing both packs is not shown to have any dedicated smoke, fire or gas detection distinct from the pack temperature and health figures already displayed on the power/energy page. Whether the venting arrangement is itself instrumented — a flow indication, a detection loop in the vent path, or nothing — is not stated by either document. [TBD] — open item 60.
2. Action — in the absence of a dedicated indication, the pilot's only available cue is the same continuous pack temperature and health reading used throughout §B.2, read for whatever departure from normal it happens to show.
3. No contactor isolation, no suppression, and no means of pilot access to the bay in flight is described — the bay sits sealed beneath the cabin floor, and nothing in either document names a crew-accessible control for it beyond what already appears on the power/energy page.

B.13 Emergency recovery system (ERS)

A guarded mechanical handle at the centre console and a separate safe-and-arm switch, both required before a firing signal can reach the recovery-system charge; a dedicated firing battery independent of the aircraft's main low-voltage distribution; a dedicated flight-deck annunciator, near the handle, for the control unit's armed state and fault. Arming and inhibit logic are tied to weight-on-wheels and height. [GIVEN]

B.13.1 Conditions in which the handle is the remaining option

There is no mechanical or hydraulic linkage anywhere between the flight deck and a control surface or propulsor (§7.1 of the body document, restated at §B.7 above). Three computing lanes carry every command electrically, and there is no mechanical reversion path behind them. Read together with §B.7.2, that means a condition removing electrical command capability from all three lanes — however such a condition would arise, and neither document names one — leaves no means of directly controlling the aircraft's flight path at all.

1. Indication — the dedicated annunciator's armed state and fault reading, and whatever handling the pilot can still feel or read through the centre inceptor and energy lever, which on this description would be nothing.

2. Action — the guarded handle, once the safe-and-arm switch permits it, is the design's stated means of addressing a flight-control condition beyond this one. No procedure precedes it, because no procedure exists for restoring command capability once it is gone; this is stated as a description of the design, not as an assessment of whether the handle is a sufficient answer to the condition that leads to it.
3. What height and weight-on-wheels combination the inhibit logic permits firing at, and what a pilot flying at low height over an elevated urban pad — as every approach and departure on this type is — can actually expect the handle to do for them at that height, is [TBD] — open item 61.

B.13.2 Inadvertent arming or inhibit disagreement

1. Indication — the dedicated annunciator shows armed state and fault; whether an inhibit condition and an armed indication can disagree with each other, and what is shown if they do, is not stated.
2. Action — the safe-and-arm switch is set according to phase of flight (Appendix A §A.4.1 step 7, §A.10 step 2); no procedure is described for resolving a disagreement between the switch's own selection and what the annunciator reports, or between the inhibit logic's weight-on-wheels and height inputs and the phase the pilot believes the aircraft to be in. [TBD] — open item 61 applies equally here, as the same control unit and the same annunciator are involved.
3. Whether the annunciator's own supply and processing path is independent of the two primary displays' failure modes is already tracked at §8.3 of the body document without a numbered item; it bears directly on this entry, because an annunciator that shares a failure mode with the displays it is meant to stand apart from would show a fault at exactly the moment its independence matters most.

B.14 Vertiport and environment

The operating concept places every sector between elevated pads among buildings, with shear and turbulence off nearby structures a normal operating condition rather than an exception (§2 of the body document), and with little open ground beneath most of the route. [GIVEN]

B.14.1 Pad unavailable on arrival

1. Indication — none specific to this condition is described. Whatever the pilot learns about a destination pad's state before or during an approach is not addressed by either document; Appendix A §A.8.1 step 1 already names the same gap for pad-specific wind, shear and obstacle data, tracked as open item 33.
2. Action — no alternate-pad or diversion procedure specific to a rejected approach is described in either document (Appendix A §A.8.1 step 3 records the same absence for an approach briefing that has nothing to brief against). [TBD] — open item 62.
3. With little open ground beneath most of the route, the pilot's available options on finding a destination pad unavailable are not the same as they would be for an aircraft that can put down on open ground if a prepared surface is not available; neither document addresses what those options are for this type.

B.14.2 Wind or turbulence beyond the declared envelope

1. Indication — no declared wind or turbulence envelope for pad operations exists in either document to be exceeded, and no display or annunciation is named for a condition defined relative to an envelope that has not yet been fixed. [TBD] — open item 63.
2. Action — none is described, for the same reason.
3. The absence is worth naming plainly because it sits at the phase where it matters most: the two- to three-minute vertical departure and approach and landing phases (§2.2 of the body document) are exactly the phases in which shear and turbulence off nearby structures are named as a normal condition, and they are the same phases carrying a transition, a building-shaded pad environment, and the full display and alerting picture at once (§10.2 of the body document).

B.15 Conditions the design does not resolve

Collected from Appendix B. Each is a place where a procedure cannot be written until a design decision is taken or an analysis is run. They are offered as a starting point for the human-factors and safety work, in the same spirit VAY-HF-0001 §11.2 already states for the items carried from the body document and Appendix A.

40. **Battery string-level status** (§B.2.1). Whether a single isolated string, within a four-string pack, is shown to the pilot as anything distinct from the pack's aggregate charge and health figure.
41. **Coolant-loop reconfiguration** (§B.2.2, §B.6.1, §B.6.2). No flight-deck control is named for selecting a coolant-loop configuration; the only available response to a rising temperature or a lost loop is a change to the flight itself.
42. **Bus-tie contactor manual control** (§B.3.1). Whether the pilot can close the tie contactor to feed one bus side's loads from the other, or whether that reconfiguration is automatic, unavailable, or something else.
43. **Low-voltage bus architecture** (§B.3.2). What the low-voltage distribution feeds, beyond the two mentions the source material makes in passing, and what its loss would mean.
44. **Insulation-monitor procedure** (§B.3.3). No threshold or crew action is defined for a declining isolation-resistance reading.
45. **Individual lift-unit-loss indication** (§B.4.1). Whether the pilot is shown which of eight lift rotors has been lost, and what hover authority remains with one gone.
46. **Grouped lift-unit loss** (§B.4.2). Whether a loss confined to one boom is distinguished from a single loss elsewhere, on any display or in the flight control system's redistribution logic.
47. **Cruise-unit-loss consequence** (§B.4.3). Asymmetry indication and any change to the transition threshold following a single cruise-unit loss.
48. **Rotor fails to stow for cruise** (§B.4.4). Whether the transition is inhibited, proceeds, or produces some other behaviour when a lift rotor has not reached the stowed, edge-on position.
49. **No crew-initiated transition abort** (§B.5.1, §B.5.2). Whether any means exists for the pilot to abort or repeat a transition sequence that does not complete, distinct from the automatic-reversion gap already carried at item 17.
50. **Computing-lane-loss annunciation** (§B.7.1, §B.7.2). No indication is confirmed to exist for the loss of one or two of the three flight-control computing lanes.
51. **Actuator jam or runaway detection** (§B.7.3). No means of distinguishing a jammed or runaway actuator from normal behaviour, on any display, is described.
52. **Reversion on loss of the single satellite or inertial source** (§B.8.1). Named in passing at §6.1 of the body document without a tracking number there; closed here.
53. **Air-data disagreement resolution** (§B.8.2). Whether the forward air data probes' outputs are compared, and what is shown if they disagree.
54. **Low-speed-sensor loss at one boom** (§B.8.3). Consequence for the hover and transition control laws, which are stated to rely on independent per-boom reporting.
55. **Partial or asymmetric downlock on a four-point gear** (§B.9). Not addressed by any procedure, and the resulting landing not characterised.
56. **In-flight door-latch disagreement** (§B.10.1). Addressed for the pre-departure check; not addressed for an indication appearing in flight.
57. **Passenger incapacitation or interference** (§B.10.2). No means beyond the open sightline over the cabin, and no procedure for what the pilot does about what they see.
58. **Pilot incapacitation** (§B.11). No detection, no second pilot, no passenger means of controlling the aircraft, and no procedure of any kind.
59. **Flight-deck smoke or fumes isolation sequence** (§B.12.1). No specific sequence, and no smoke-removal procedure distinct from isolation and landing, is described, and no oxygen equipment exists to reach for while performing it.
60. **Battery-bay detection and indication** (§B.12.2). Whether the sealed, vented bay carries any dedicated detection beyond the pack temperature and health figures already on the power/energy page.
61. **Emergency-recovery firing conditions at low height** (§B.13.1, §B.13.2). What the arming and inhibit logic's weight-on-wheels and height inputs actually permit at the heights this type's urban pad approaches and departures are flown at, and whether an inhibit condition and an armed indication can disagree.
62. **No diversion procedure for a pad unavailable on arrival** (§B.14.1). Pad-specific condition data is named at open item 33; the diversion procedure itself is a separate, equally open gap.
63. **No declared wind or turbulence envelope for pad operations** (§B.14.2). Nothing exists yet to be exceeded, and no procedure exists for recognising or responding to exceeding it once it does.

Two of these are set apart because they are not, in the end, gaps of the same kind as the rest.

Pilot incapacitation (item 58) is not a case where a procedure is missing; it is a case where nothing in the aircraft's description — detection, a second seat, a passenger means of control — gives a procedure anywhere to attach to. Every other item in this list describes an indication or an action the design could plausibly add. This one describes an occupant the design does not have.

The condition behind the emergency recovery handle (item 61, read together with §B.7.2 and §B.13.1) is the item this appendix would most like the safety and human-factors work to take up first. A design with three electrical computing lanes and no mechanical reversion of any kind has made a choice: when the electrical command path is gone, nothing remains between the pilot and the guarded handle. That choice is stated here as a design fact, not as a judgement on it either way. What this appendix can say is that the aircraft flies every sector low, over buildings, with little open ground beneath it, and that the arming and inhibit logic deciding whether the handle does anything useful at the height and configuration the aircraft is actually in when it might be needed is exactly the piece of the design this appendix cannot characterise, because neither source document gives it anything to characterise it with.

Appendix C — Control schedule

Every control or dedicated indication named anywhere in VAY-HF-0001 §1–§11, in Appendix A, or in Appendix B above appears here once, grouped by the panel-zone table at body document §4.2. Where VAY-SDD-0001 places the control or fixes its function, the row is marked [GIVEN]; where its position or form is this document’s own construction, built from a function the source material names without naming the control itself, the row is marked [PRELIM]; where neither the control’s existence nor its function is stated at all, the row is marked [TBD]. This schedule does not fix breakout force, control travel, detent force, or actuation direction — those are open items 11 and 12, and they apply to every row below without exception. Guarding provisions are marked where VAY-SDD-0001 states one; where it does not, the guard is [PRELIM] or [TBD] as the row indicates, and open item 13 applies throughout.

This schedule is shorter than a comparable two-crew transport’s control schedule would be, and that is worth stating plainly rather than leaving the reader to wonder whether rows were left out. It is not shorter because the aircraft has fewer functions to manage — ten propulsors, two battery packs, two bus sides, two coolant loops and a fly-by-wire system across three computing lanes are, if anything, more to manage than a comparable conventionally-propelled aircraft carries. It is shorter because VAY-SDD-0001 names far fewer of those functions at flight-deck-control granularity. Several rows below record a control this document believes must exist, because the function it would serve is named somewhere in the source material, without VAY-SDD-0001 naming the control itself, its panel, or its form. Those rows are marked [TBD] rather than omitted, on the same principle §1.1 of the body document states for every other gap in this document family: an empty field, honestly marked, is the correct answer.

Tag	Legend	Panel	Type	Function and provenance
1	MASTER	ALERTING PANEL	pushbutton annunciator	One of the two glareshield alert lights. Alert messages appear on whichever display shows a synoptic or status page, with an aural tone and one of these lights; which light accompanies which message is [TBD]
2	MASTER	ALERTING PANEL	pushbutton annunciator	Marked <i>CAUTION</i> . As above. [GIVEN]
3	ALERT	ALERTING PANEL	pushbutton	Marked <i>ACK</i> . Glareshield acknowledgement switch; the pilot can then call up the associated system page. Whether that page carries the alert’s procedure or only its status is [TBD] — open item 28. [GIVEN]
4	ERS	DEDICATED ANNUNCIATOR	indication only	Marked <i>ARM / FAULT</i> . Emergency recovery control-unit status, presented on “a dedicated flight deck annunciator near the activation handle” rather than on a display page. Independence of its supply and processing path from the two primary displays’ own failure modes is [TBD] (§8.3 of the body document). [GIVEN]

Tag	Legend	Panel	Type	Function and provenance
5	DU L	MAIN INSTRUMENT PANEL	display surface	Left display, own graphics processor; configurable to flight, navigation or synoptic pages. [GIVEN]
6	DU R	MAIN INSTRUMENT PANEL	display surface	Right display, as left. Either surface can show either page. [GIVEN]
7	BEZEL L	MAIN INSTRUMENT PANEL	bezel key set	Page selection for the left display; manual throughout, no stated default per phase — open item 8. [GIVEN] as to manual selection
8	BEZEL R	MAIN INSTRUMENT PANEL	bezel key set	As left. [GIVEN]
9	STBY	MAIN INSTRUMENT PANEL	standby display	Attitude, airspeed and altitude equivalents. Architecture, and independence of its supply, sensor and processing path from the two primary displays, is [TBD] — open item 10. [GIVEN] as to existence
10	AUDIO PANEL	MAIN INSTRUMENT PANEL	control panel	Selects and routes COM, alert tone and, where fitted, intercom audio to the single headset. Arbitration between an alert tone and radio or intercom audio on the shared path is [TBD]. [GIVEN] as to existence and the shared path
11	GEAR	LOWER INSTRUMENT PANEL	hard switch	Landing gear control. Hard switches for gear are described at the lower panel rather than overhead. [GIVEN] as to panel location; switch form [PRELIM]
12	LIGHTING	LOWER INSTRUMENT PANEL	hard switch(es)	System lighting control(s); grouping and count not detailed. [PRELIM]
13	SYS RESET	LOWER INSTRUMENT PANEL	hard switch(es)	System reset switches, named generically; which systems and how many resets is not detailed. [PRELIM]
14	GEAR ALT EXTEND	LOWER PANEL / CENTRE CONSOLE	pull handle	Manual free-fall extension, from a separate handle, independent of the electric gear actuators. [GIVEN] as to the handle and its independence; panel position [PRELIM]
15	INCEPTOR	CENTRE CONSOLE	centre inceptor	Single centre inceptor; flight-path command, with two independent signal paths into the flight control system. [GIVEN]
16	ENERGY LEVER	CENTRE CONSOLE	lever	Single left-hand energy lever; power command distributed across all ten propulsors by the flight control system, with two independent signal paths. [GIVEN]
17	ERS	CENTRE CONSOLE	guarded pull handle	Marked <i>PULL TO DEPLOY</i> . Emergency recovery system activation handle. Guarded; firing requires this handle and the separate safe-and-arm switch together. [GIVEN]

Tag	Legend	Panel	Type	Function and provenance
18	ERS	CENTRE CONSOLE	guarded toggle	Marked <i>SAFE / ARM</i> . Separate safe-and-arm switch, set according to phase of flight. Guard provision beyond the handle's own guard is [PRELIM]. [GIVEN] as to the switch and its function
19	DOOR	CABIN/DOOR — panel not named	switch, form not stated	Marked <i>RELEASE</i> . Commands door-latch release once the aircraft is confirmed at the pad. Neither the panel nor the control's form is named in either source document. [TBD] — open item 64
20	DOOR L	CABIN/DOOR	manual release lever	Left door manual release; ground use; independent of the electric latch actuator. [GIVEN]
21	DOOR R	CABIN/DOOR	manual release lever	Right door, as left. [GIVEN]
22	CABIN LTG	CABIN/DOOR — panel not named	control, form not stated	Cabin lighting: two strings, one per sidewall, on separate low-voltage feed segments, with ceiling trim lighting providing overlap. Adjustable by the pilot from the flight deck as a whole; a control for selecting or overriding either string individually is not named. [TBD] — open item 65
23	ENV OUTLETS	CABIN/DOOR	control, form not stated	Forward/aft environmental outlets, adjustable by the pilot from the flight deck. [GIVEN] as to adjustability; control form [PRELIM]
24	VHF 1	COMMUNICATIONS — panel not named	radio control head	First VHF radio. [GIVEN] as to existence; panel and control form [PRELIM]
25	VHF 2	COMMUNICATIONS	radio control head	Second VHF radio. [GIVEN]/[PRELIM], as above
26	DATALINK	COMMUNICATIONS	control/display, form not stated	Vertiport datalink transceiver. [GIVEN]/[PRELIM], as above
27	PA	COMMUNICATIONS	microphone / switch	Marked <i>OVERRIDE</i> . Pilot override of the automatic, phase-triggered passenger-address announcement, from the flight-deck microphone. [GIVEN]
28	BUS TIE	ENERGY/PROPULSION — panel not named	contactor control, form not stated	Manual selection, if any, for the high-voltage bus-tie contactor. Not named as a crew-accessible control anywhere in VAY-SDD-0001. [TBD] — open item 42
29	COOLANT	ENERGY/PROPULSION	control, form not stated	Marked <i>LOOP CONFIG</i> . Coolant-loop configuration selection, if any. Not named as a crew-accessible control. [TBD] — open item 41
30	DATA BUS	ENERGY/PROPULSION or LOWER PANEL	control, form not stated	Marked <i>RESET</i> . Resetting a data-bus channel, named as a systems function in §7.2 of the body document without a stated flight-deck control. [TBD] — open item 66
31	GND CHARGE	ENERGY/PROPULSION	control, form not stated	Ground/network charging initiation, if pilot-initiated as distinct from automatic once ground equipment connects. [TBD] — open item 30 (Appendix A)

Tag	Legend	Panel	Type	Function and provenance
32	FLT MGT	MAIN INSTRUMENT PANEL (assumed)	control, form not stated	Route loading and flight-management data entry. Existence implied by Appendix A §A.4.1 step 11; means of entry not detailed. [TBD]
33	—	OVERHEAD PANEL	zone, contents unconfirmed	Whether the type has an overhead panel at all, distinct from one simply not yet described, is [TBD] — open item 67. No control below is placed there because none can be, honestly, until the zone itself is confirmed.
34	—	SIDE CONSOLES	zone, contents unconfirmed	As above. [TBD] — open item 67

Rows 33 and 34 are included as zones rather than as controls because §4.2 of the body document names them as [TBD] in exactly that form — a panel zone that may or may not exist, as distinct from a control whose panel is merely unassigned. Every other row above at least has a function to anchor it, even where its panel or form does not yet exist; these two rows do not have even that, and are carried here rather than dropped so that a later assessment does not have to rediscover that the question was ever open.

Four items are worth drawing together across the table rather than leaving them scattered row by row. First, no control in this schedule has a second instance for a second crew member to operate, check, or reach for independently — every row above is reached, read and actioned from the one centreline seat, and Appendix A §A.2 already states what that means for how each control’s use is written. Second, several rows exist in this schedule only because a function is named without a control being named to go with it — rows 19, 22, 28, 29 and 30 are each a case where VAY-SDD-0001 states plainly that something happens (a door releases, lighting adjusts, a contactor opens, a coolant loop reconfigures, a data-bus channel resets) without stating how the pilot makes it happen, if the pilot makes it happen at all. Third, the two zones at rows 33 and 34 mean this schedule cannot yet claim to be complete in the way the Aeolus schedule this document was built against could: that schedule drew every control from a stated flight-deck layout; this one draws its rows from what sixteen system descriptions happen to mention in passing, and a panel zone neither document confirms or denies is not evidence of an empty panel, only of an unfinished description. Fourth, and following directly from the first three, the guarded controls in this schedule — rows 4, 17 and 18 — are also the ones §B.13 above depends on most directly, and nothing in this table adds to what §B.13 already says about what the guard protects against or what lies beyond it once the guard is opened.

The Vayu VY-6 is a fictional aircraft. This document demonstrates the shape and the discipline of a flight-deck human-factors input specification. It is not a design of, or a statement about, any real aircraft.